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Mind-Oriented and Virtue-Oriented Learning Processes

Given the cultural learning models, an important question to consider is how these cultural models may influence the *actual* learning process in which the members of each culture engage. I attempt to address this question in this chapter. Before proceeding, however, it is important to lay out different kinds of human learning and clarify to what kind of learning the cultural models are most relevant.

KINDS OF HUMAN LEARNING

It is a well-established scientific fact that human beings are capable of learning virtually unlimited things. Human learning takes place in many ways, and it starts even before birth. For the most part, infants and young children learn much about the world without being specifically taught; for example, they acquire vocabulary of their native tongue and social norms of behavior without deliberate adult instructions.[1] However, because it proceeds more or less without the need for effort on the part of young children or deliberate instructions on the part of adults, this kind of individual learning is not directly influenced by culture. It is instead testimony to the human capacity to learn.[2]

However, the emergence of human culture led to the accumulation of knowledge and skill, including all the tangible and intangible cultural artifacts (e.g., tools and symbols). This cultural development demanded a totally different kind of learning, one that requires effort from the learner and effort from the person who can impart the knowledge/skill. This kind of learning can still be seen widely in informal settings in today's world, such as a mother teaching her child how to feed pigs, or an older brother showing his younger siblings how to shoot a jump shot.

As knowledge and skills continued to accumulate and division of labor widened during the long preliterate human history, most cultures relied on *apprenticeship* as a form of learning. In this system, most learners went to a master to learn a craft or trade such as blacksmithing or pottery making. Apprenticeship learning took an extended period of time to complete before the apprentice could be regarded as having mastered the trade. The typical learning process was one-on-one teaching, with the apprentice observing the master intently and starting with low-levels of tasks and gradually taking on higher levels of tasks. Most skills were not explicitly explained by the master but were embedded in the activities. It was the apprentice's responsibility to learn the skill. Apprenticeship learning was also part of productive labor, producing tangible products such as tailoring and carpentry that contributed to the livelihood or the business of the master.[3]

While apprenticeship learning continued, human cultures also developed literacy, arithmetic, and other higher-level and specialized knowledge/skills. Such new forms of knowledge/skill enabled better human productivity and survival; therefore, transmitting such knowledge/skill became necessary. These new forms of knowledge/skill differed fundamentally from previous forms in that they could be written down on material (clay, stone, bronze, bark, cloth, and later paper) for preservation, unlike the previous reliance on human memory and oral transmission. This advancement of knowledge preservation also eased transmission of knowledge. This development ushered in formal schooling where such knowledge could be systematically transmitted to the young. The earliest formal schooling begun in the Middle East around 4000 BCE and in China during the Xia Dynasty around 2000 BCE, where young boys were taught literacy and other important skills (poetry, music, and rites, among others). However, before modern time, formal schooling existed only for the ruling class and the elite in the West and aristocracy in China.[4]

The learning that formal schooling demanded was yet a shift in kind. Reading, writing, and arithmetic lay in the heart of schooling, and they were encoded in intangible symbols that were abstract by nature. Learning such things was different not only with regard to content, but more importantly to the great *mental* effort that was required from the learner. Yet, learner's effort alone was no guarantee for mastery; a specialist, the teacher, was needed for ensuring such learning. The teacher had to achieve the intended knowledge in the first place, usually demonstrating superior mastery before he could be acknowledged and granted the privilege to teach the young.

When teaching, the teacher had to continue to exert sustained mental effort to ensure his pupils' successful learning.

With industrialization came mass education. For the first time in human history, formal schooling was not controlled by the powerful and the elite but extended to every child wherever the compulsory education law exists. Compulsory education is still expanding, and it is quite possible that such education will be realized across the globe in the foreseeable future.[5]

As knowledge grew more extensive, it also grew more abstract. Both the quantity and increasingly abstract nature posed great challenges to learning. How best to teach and to learn given the specific cultural heritage became important questions for many cultures. Western and Chinese/East Asian learning traditions were developed in response to these questions and concerns. As discussed in Chapter 2, the different paths that the West and East Asia followed resulted in quite different approaches to learning.

However, present-day formal schooling shares some common characteristics regardless of the culture in which education takes place. Two such commonalities are particularly noteworthy. First, the world has been witnessing convergence of knowledge, mostly because of Western contributions. Such knowledge has become the common human heritage regarded as important for any child to learn. Mathematics and science are such knowledge. There is hardly any current formal education system in any culture that does not include math and science in their curriculum. Second, formal schooling proceeds uniformly inside the classroom with additional learning assigned outside the classroom. This kind of learning aims at mastery in the most efficient way by the greatest number of learners. Regardless of how much inherent ability an individual learner has, any given learner is likely to spend much more time and to exert much more effort to acquire currently desirable knowledge than the same learner would have a century or two ago, regardless of the culture.

Still, as research documents, individual culture's ways of learning (and teaching) as formed in their long histories are unlikely to disappear even for cultures that have incorporated other cultures' bodies of knowledge and pedagogy. Instead, these culturally formed ways of learning may continue to shape how learners actually engage in their learning. What seems to emerge is that cultures retain the best of their own practices as a more basic learning approach while assimilating elements from other cultures in order to respond to new challenges of learning. It is therefore important to describe each culture's basic learning process. This understanding could help us appreciate the more complex process of borrowing and incorporation of new elements from other cultures.

LEARNING PROCESS IN WESTERN AND CHINESE CULTURES

In this section, I describe how each culture's learners typically engage in the learning process. These portrayals of the respective learning processes are based on empirical research on learners' self-descriptions and reflections, observations of academic learning inside and outside school, as well as observations of other but equally effortful types of learning such as painting, dance, martial arts, and playing musical instruments. Although some examples come from specific individuals, particularly the cultural icons of learning, I do not suggest that these portrayals represent the learning processes that *every* individual of the culture follows. The discussion is only meaningful with regard to general cultural trends and patterns as found by research, not processes that are undertaken by specific individuals. Neither are these portrayals about pedagogy and teaching – that is, how school curriculum is organized and executed and how teachers teach in the classroom. These portrayals are about learning.

Mind-Oriented Learning Process in the West

In describing Western learners' process of learning, I highlight the processes of active engagement, exploration and inquiry, thinking and critical thinking, and self-expression and communication (see Figure 4.1).

Active Engagement

From my study of the English lexicon of learning,[6] active learning emerged as the very first major category of specific learning processes.[7] It has the following typical words and phrases: *active learning, acquainting/familiarizing oneself with something, competition, experience, getting the hang of it, hands-on, learn by doing, learning it the hard way, practice, study, training, trial and error, and work*. As can be seen clearly, these are the actual activities in which learners engage. They are behaviors that can also be observed. What is emphasized in these expressions is the active nature of learning, not what is being learned or even what strategies are being used. When learners are said to be actively engaged, they are physically present, frequently with their hands and bodies, not just their minds, fully involved.

Descriptions of European-American ideal but real learners that I collected also confirm the active nature of learning. Many respondents wrote of active involvement in the learning process, for example, reading all kinds of books and newspapers, browsing the Internet, writing papers, performing observations, trying out ideas, and getting involved in activities.

Aside from these words and descriptions, it is a common scene in most Western schools that learners are highly encouraged to go to the library and get on the Internet to research topics of interest, to read books, to write essays and reports of their discoveries, to construct objects (e.g., a papier-mâché model of a volcano), to make models to demonstrate their understanding (e.g., a cardboard box showing the terrain and animals of the tundra), to do scientific experiments (e.g., lab experiment or following a hurricane's movement over time), and to document their observations.

Another common form of active learning is taking children out on fieldtrips. Entire classes of students go to real-life settings such as museums, aquariums, observatories, historical sites, power plants, and factories. Students take notes on site and discuss and write about what they have learned. A third form of active learning is the kaleidoscopic afterschool program. Many public and private schools offer many different kinds of programs that are, by nature, learning by doing (activities). Children learn crafts, such as how to draw, sculpt, play musical instruments, make books, make jewelry, dance, and play various sports. Finally, during the summer break, students also join camps and organized trips to other places within their country and abroad. Many such camps and trips are for fun, but many have a strong component of learning by doing.

The best cases can be found in many Western preschools such as Montessori schools, schools inspired by John Dewey's philosophy such as Shady Hill School in Cambridge, Massachusetts and Putney School in Vermont, and without doubt the much admired preschools of Reggio Emilia, Italy. What strikes visitors not from the West (and even from the West) is how much emphasis is placed on the active nature of children as learners. Sitting in a classroom to be instructed day in and day out is not the mode of these exemplary learners. They engage in learning with their senses, curiosities, experiences, and questions about their surroundings, human or physical, real or imagined. They draw, build, investigate, and design things in order to learn and to study them. For example, stepping into a puddle after rain leads to children's study of how water and mirrors reflect things. Children draw their images standing by the puddle or on top of a mirror, learning and thinking about what they discovered. These young learners also study man-made modern environments such as supermarkets by gathering information, by talking to store managers and shoppers, and by making suggestions on how to improve the stores.[8]

Children also have plenty of opportunities to be active in learning outside school. Many museums are interactive just for that purpose. In other words, children are welcome to visit museums where they can touch things, take things apart and put them together, play, experiment, and explore. In Philadelphia, there is even a children's museum called the "Please Touch Museum" where young children are invited to start touching and playing with their exhibits as soon as they enter the museum. Children's museums in the West are exemplary in their effort to actively involve children in learning and exploration.

In addition to and oftentimes embedded in the observable active learning engagements, there are a host of what is called *learning strategies* or *self-regulated learning strategies* that Western learners employ in their learning. These may not always be observable, but they are nonetheless very active processes. In general, learners use three kinds of strategies: cognitive strategies, metacognitive strategies, and support strategies as Paul Pintrich and colleagues labeled them. The first kind, cognitive strategies, are more basic. They include three specific processes: (1) processing of information from written material and lectures such as rehearsal (e.g., repeating words over and over and using flash cards to recall information); (2) elaboration such as paraphrasing concepts and summarizing readings; and (3) organization such as outlining, grouping, and combining previously unorganized material. The second kind, metacognitive strategies, also include three specific processes: (1) planning (e.g., setting goals and steps), (2) monitoring (e.g., one's own progress and understanding), and (3) regulating (e.g., allotting and adjusting one's review according to one's understanding). The third kind, resource strategies, include (1) one's time management (e.g., prioritizing one's competing tasks), (2) maintaining a conducive study environment, (3) regulating one's effort (e.g., increasing effort for more challenging work), (4) peer learning (e.g., forming study groups), and (5) seeking help when one needs it. These strategies have indeed been found to be commonly used by Western learners (more than Asian).[9]

Exploration and Inquiry

As detailed in Chapter 2, inquiry is of central importance to Western learning. What then does the learner do when he or she is said to engage in the process of inquiry? The learning terms that were sorted into the conceptual group *inquiry* in my lexicon study contained these typical expressions: *visual intake*, *visualization*, *questioning*, *brainstorming*, *exploring*, *breaking things down*, *experimenting*, *applying ideas*, *relating ideas*, *analyzing*, *problem* *solving*, *putting things together*, and *synthesizing*. These terms emphasize the mental activities in the process of inquiry that learners carry out.

There are many facets of the notion of inquiry, and all are important. First, our most common human experience in daily life enables us to notice discrepancies between our current beliefs or knowledge about things in the world and those of others, or discrepancy between our self-derived beliefs and those of authority (i.e., parents, teachers, scientists, politicians, media, and government). For example, as a case for the first kind of discrepancy, a child has learned how to swim and believes that everyone can learn how to swim, but her friend tried to learn and failed. The friend told her that she cannot learn how to swim. The child is likely to be in a state of realizing that her own belief and her friend's are different. As an example of the second kind of discrepancy, an elementary school student thought that lobsters were gigantic bugs when his parents first brought them home. He therefore did not believe that lobsters were edible, and in fact the idea of eating bugs was scary and turned him off. But his parents kept inciting him into tasting the delicious "bug." If the child were an inquisitive learner, he would likely wonder why a bug could be delicious as claimed by his parents, and perhaps he would want to find out why his parents did not think lobsters were bugs.

In situations like this, most people raise questions. When Western learners find themselves in this state, they are not only admired but highly encouraged to express their questions (not just keep the questions to themselves) and to look for answers on their own. Thus, being observant about existing beliefs and knowledge in oneself and in others and recognizing discrepancies are the first step in order for one to raise questions. Questions carrying the potential of prompting an inquiry are necessary before the learner can continue. This is why in Western intellectual tradition and pedagogy, asking questions is so prized. Being able to pose good research questions is the high point for graduate training, dissertation writing, and ultimately a successful academic life.

Curiosity and interest are similar mental states conducive to asking questions. A learner who is not curious and not interested in a topic is not believed to be able to raise questions, certainly not good and worthy questions for inquiry. When a learner is curious about something, he or she wants to know more about it. When a learner is interested in something, he or she moves beyond the momentary spurt of attention to which curiosity may be closely linked. For example, a learner with an interest in math is someone who likes math and is eager to learn more on that subject. The fine distinction between curiosity and interest in motivating the learner is not the focus here (the affective dimensions will be discussed in Chapter 5).[10] Suffice it to say that both of these mental activities and attentional states are likely to generate questions similar to noticing discrepancies. Curiosity and interest frequently do result in good and worthy questions that can lead to inquiry. In Western learning, curious and interested learners are in some ways more prized than the discrepancy-dependent questioners because their curiosity and interest are indicative of an enduring and highly valued disposition in the learners: inquisitiveness.

After raising questions, the learner may begin the process of *exploration*. This process for children, particularly toddlers and preschool children, is usually, barring any endangerment, unrestricted. Children are ensured freedom to explore anyway they prefer, be it something that happens to draw their attention or something that is their "obsession" such as a child being completely enamored with dinosaurs. Adults and teachers are deliberately hands-off, leaving children to wander off, to take apart or put together objects (and many do break objects), to ask questions endlessly, to make noise, even to make mistakes and get frustrated. Adults and teachers are reluctant to spoon-feed children with the "right" ways to do things or the "correct" answers to their questions because they believe that children's natural curiosity and interest need to be protected and promoted. Overly involved adult teaching in this process is considered as "interfering" and therefore stifling children's sense of wonder about the world, which is regarded as the most precious quality in children.

Spontaneity in children's learning is synonymous with their exploration. Children, particularly preschool children, are believed to possess a spontaneous thirst for learning. This belief rests largely on the scientific fact that children's rapidly developing brain and body enable them to perform many cognitive and social feats such as inventing words and other symbols and to imagine social scenes in their pretend play without the apparent need for heavy-handed instructions from adults. Preschool children are also admired for their innocence from conventional constraints. This is one reason why children's thoughts, expressions, drawings, and behaviors seem inherently creative to adults.[11] Some world-renowned artists – for example, Picasso, Klee, and Miro – were so fascinated with children's drawings that they imitated them and perhaps discovered their essence. Constraining young children in the classroom and subjecting them to the strong discipline of studying are considered inappropriate because this type of adult control makes children passive. Passivity in children is the death of exploration and inquiry. Passivity, therefore, is to be prevented as much as possible.

If a preschool in the West wants to be successful, the best way to achieve that goal is to make children spontaneous, lively, and exploratory learners.

A vivid example offered by Howard Gardner in his book *To Open Minds* can illustrate this precious process of exploring by their toddler son Benjamin and their strong effort to protect the child from being "interfered" with by well-intentioned but not so inquiry-oriented Chinese adult helpers:

> When Howard Gardner and his wife Ellen Winner spent a month in Nanjing in the late 1980's for studying Chinese children's arts education, they stayed in a hotel. There was a key slot for guests to insert their keys before leaving the hotel. Their one-year-and-half-old son Benjamin enjoyed playing with the key. He also tried to put the key into the key slot. But because the slot was narrow and rectangular, it was hard for the toddler to succeed. Since he seemed to have so much fun with this key and the key slot, the parents did not intervene, leaving him alone to engage in his exploration. But the Chinese hotel attendants and passers-by would watch the child, then show and even help him put the key into the key slot.[12]

One repeated reason that the preschools in Reggio Emilia are regarded as the champion of Western preschool education is that children are truly allowed to explore themselves and their worlds. These Italian children enjoy complete freedom of exploration. But unlike the incident described in Gardner's book, these children's explorations – including their movements, activities, questions, conversations, and other utterances – as well as their detours and setbacks are recorded, analyzed, and revisited. In other words, their explorations are valued so much that the skillful teachers actively follow up, support, and guide these children's explorations to the next level, bearing intellectual and artistic fruit. These children's drawings, sculptures, and other constructed objects as well as their conversations with peers stun the world with their spontaneity, sophistication, and beauty, yet show no trace of adult educators' interventions and impositions. Reggio Emilia once again demonstrates that there is no limit to how much children can explore and inquire even at preschool age.[13]

The inquiry process involving older children and college students is somewhat different. Older learners, too, are encouraged to explore and ask questions, but the actual process is not purely self-initiated and spontaneous with parents and teachers staying hands-off. School-aged learners attend compulsory education in most Western countries. This means that they must receive either a certain number of years of education (typically nine or more years) or they must attend school up to a certain age (e.g., sixteen in most U.S. states). Schools are usually charged to teach a curriculum of government-mandated content (e.g., subject matters), pedagogy (e.g., there must be instructions and feedback given to students), and achievement standards (e.g., passing statewide or national tests). Completely free exploring by students alone cannot guarantee the attainment of such a curriculum.

However, compared to Asian learners, Western schoolchildren still have much more freedom to inquire into many topics. Beyond sitting in the classroom listening to the teacher, children are often asked to build models. For instance, in learning about atoms, middle school children may be asked to learn the basic structure of atoms, and then go home and find household materials (e.g., yarn, cloth, dried noodles, plastic, and crayons) to build a model of an atom. Each child then brings a model built with different material back to school. This is a typical process of engaging schoolchildren in the process of inquiry. Without such personal exploration, children may not be able to gain such clear understanding of what atoms are and how they function. The diverse approaches by different children also serve to inform each of them that there are many ways to achieve understanding and discoveries.

Children also frequently replicate scientific experiments in class that were done by the original scientists who discovered the laws. By going through the experiments, children learn step by step how such laws were discovered. Oftentimes, children are given opportunities to vary from the original experiments in order to formulate their own hypotheses, gather genuinely new data, and draw informed conclusions. In other words, this learning process models after the real scientific inquiry. Moreover, children participate in science fairs in most schools, where they must come up with their *own* research topics without any teacher input and follow the inquiry process to complete their projects.[14] At the science fairs, they present their discoveries and the steps they took to make their discoveries.

In English and social studies, inquiry is also highly valued. Children's learning in these disciplines is not characterized by memorization, for example, of a whole model essay or poems in order for them to imitate. Instead, they are required to investigate and analyze topics. For investigation in a history class, for example, children may dress up as historical figures in order to reenact an important event, or they may build a set and present a short scene about a historical movement. Children may also interview real people who lived through the Great Depression or World War II and then write a research report about what they found. My son in third grade engaged in his English project where he was asked to imagine himself as a plant. His task was to observe and record how he-as-the-plant grew, what support he needed, what enabled him to mature, and what maturity entailed. He went to his grandmother's vegetable garden behind their house and found a bitter squash sprout. He decided to be that sprout. He went out everyday and observed how the sprout grew. He also watered the plant. He took notes on all of his observations. Eventually, this plant bore many bitter squashes growing from small to large. He wrote an essay about the sprout's growth and its life stages. In this type of learning, inquiry as inspired by great scientists such as Darwin is still commonly promoted and prized in Western schools.

This exploratory and inquiring spirit is not limited just to school, but it also permeates the whole society, particularly children's TV programs and Web sites. As an example, in the United States, Public Broadcasting Services (PBS) presents preschool and school-aged children with many excellent daily shows that focus on engaging children in exploring and inquiring into the world, ranging from the jungle, the ocean, and space to the human world of cities, countryside, neighborhood, and home. There are also shows that teach children and teenagers how to design things and how to find out about things in the world. Among these shows, *Curious George*, *It's a Big, Big World, Kratt's Creatures*, and *Design Squad* are children's favorites. Similarly, many commercial TV programs also offer shows that encourage children to explore the world.

Finally, the Nobel Museum in Stockholm keeps the autobiographies of Nobel laureates. A perusal of recent Nobel Laureates in physics reveals clearly that all of these scientific icons, almost without exception, spent their childhood exploring, investigating, building things, and tasting the fruits of their own discoveries. Here is the testimony of George F. Smoot, who won the Nobel Prize in physics in 2006 for his co-discovery of the blackbody form and anisotropy of the cosmic microwave background radiation. As a child, he was curious about the moon:

> The first pivotal event I remember in my development as an inquiring scientist began with a visit to my cousins, as we took a night drive across the state of Alabama.… Excited from the visit with my cousins, I stayed awake, looking out the window rather than taking a nap. I noticed that the moon was following us mile after mile tagging along like my dog but with greater speed and persistence. I asked my parents, "How does the moon know to follow us?" They told me that the moon followed all the cars, not just ours. I was intrigued and wanted to know how the moon did this. Was it immensely superior to a dog? My parents patiently explained that the moon was very big and very far away and thus the angle did not change noticeably as we drove for miles. They gave examples of near and

> far objects that we could see along the way, so I could realize this for myself. I was impressed by how big and far away the moon must be but even more impressed that one could understand what they saw in the world and that it was so beautifully simple and clear when visualized in the proper way. It was a startling revelation that the world could be understood by simple rational evaluation.[15]

Roy J. Glauber, the 2005 Nobel laureate in physics for his contribution to the quantum theory of optical coherence, also wrote about his childhood curiosity and exploration:

> [W]hen I was four, we actually had a radio. It occupied a wooden cabinet about the size of a steamer trunk. I remember insisting there must be a man inside it. He had given his name as Maurice Chevalier. Discovering that the cabinet top was hinged, I opened it and can still feel my bafflement at discovering within it only a few glowing radio tubes. Electricity mystified me throughout childhood and I vividly remember once at age seven trying to see what it was all about. Plugging lamp cords into wall sockets must lead to the flow of something through those wires, but whatever it was, one never got to see it before it was swallowed up by the lamp. One morning I awoke early, determined to catch sight of it. I screwed the wires of a short length of lamp cord into a male plug and inserted it into a wall socket, leaving free the frayed wires at the other end. There was a bright blue flash at that end, accompanied by a muffled bang. That was followed by silence, till my parents awoke and began wondering why none of the light switches seemed to be working. The fuse was easily replaced, but I never overcame my surprise that what passed so silently through slender wires could behave so aggressively.[16]

Although these scientists received great encouragement and support from their parents or teachers, it is clear that their wonders about the world and their inquiring passion since their childhood have been the real drive for their later achievements.

Thinking and Critical Thinking

As discussed in Chapters 2 and 3, Western intellectual tradition places heavy emphasis on the mind, and one great power of the mind is to think. From my lexicon study on learning concepts, the following terms were rated as highly relevant to learning and sorted into the *thinking* category: *thinking*, *critical thinking*, *reasoning*, *derivation*, *deductive*, *inductive*, *infer*, *free thinking*, *absorbing*, *challenge*, *challenging assumptions*, *contemplating*, *pondering*, *enlighten*, *expanding*, *internalize*, *introspection*, *realize*, *recognition*, *self-reflection*, *discovery*, and *understand*. This array of thinking terms refers to different aspects, processes, and functions of thinking. Profiles of ideal learners also contain many descriptions of all kinds of thinking processes.

There are three levels of thinking: (1) pure mental processes, (2) thinking geared toward understanding, and (3) critical thinking. With regard to pure mental processes, the Western learner engages in reasoning, which includes more specific steps such as breaking things down, separating things, using logic (deductive reasoning), inductive reasoning, comparing, contrasting, and constructing things, as well as more general processes such as imagining, making connections, and synthesizing. For example, when learning how a particular plant grows, the learner may try to think about what the plant needs in order to grow: water, sunlight, and nutrients. The learner can then explore if these things are supplied to the plant in a sequence or simultaneously. After these initial specific thinking processes, the learner can then compare and contrast this plant with a plant that requires a different amount of water and sunlight and a different kind of soil and nutrients, to generate fuller knowledge of the first plant.

For achieving understanding, the learner needs not only to engage in the specific processes of thinking to master the knowledge under study, but also to go beyond. As David Perkins at Harvard Project Zero has advocated, learners must be able to explain a given concept or a theory learned from a lesson/lecture to themselves and to others with telling examples. Being able to explain something to someone else is a common way to demonstrate understanding. But learners do not stop there; they further try to generate hypotheses in order to predict related phenomena. Finally, they continue to extend their knowledge to similar situations either to verify their knowledge or to generate new ways to think about the original topic under study.[17] For example, when an elementary school student has understood how fractions work, he or she should be able to explain the concept of fraction and demonstrate to others how fractions are related to decimals and how the two can convert into each other. Moreover, when given new problems of fractions, the student can solve them and even use fractions to compute real-life problems such as comparing unit prices for products at a grocery store.

Many European-American respondents in my ideal learner study also wrote about these thinking processes. For instance, in the words of a respondent who provided descriptions of his ideal learner, a great learner recognizes "situations where information theory is an appropriate method and where it is not; [he] … performs calculations and knows why the specific steps of such theoretic calculations are as they are; [and he] understands how information theory can be applied to various issues."

The third level of thinking in which learners engage is critical thinking. As stated in Chapter 2, critical thinking is a highly valued learning process as well as outcome in the West. Many educational psychologists regard critical thinking as the cultivated personal disposition that marks the pivotal achievement of a learner. Although this educational goal has not been widely realized for all learners, it remains as a high ideal in the Western educational agenda, which continues to inspire Western parents, schools, and faculty in higher education in their effort to instill this ideal in their learners.[18]

Critical thinking is defined generally as "reasonable and reflective thinking that is focused on deciding what to believe or do."[19] More recent discussions point to the four key components that are of particular relevance to critical thinking for our purposes. The first is truth seeking, a process that leads the learner to seek the best knowledge in a given context. When engaged in this process, the learner casts doubts about existing knowledge and asks questions. Typical of Western learners is their much admired act of challenging authority. In the pursuit of truth and in exercising intellectual honesty and objectivity, the learner is encouraged to pose pointed questions and demand like responses. Similarly, the learners themselves also embrace, ideally, others' challenges to their own thoughts even when the results do not support their self-interests or preconceived ideas.

The second component is open-mindedness, which is the process that keeps the learner tolerant toward different views and propels the learner to examine his or her own possible bias. The third component is analytical process that engages the learner in examining rigorously available information that is obtained from seeking truth (e.g., gathering empirical data). A key step is using the skill and exercising the disposition of weighing the evidence against competing information and personal preferences. The learner must control his or her personal likes and dislikes and honor facts. This process is often linked to the process of problem solving, where the learner is sensitized to conceptual and practical pitfalls and to the need to revise his or her solution proposals. The fourth component is the same inquiring process as noted previously where the learner pursues personal curiosity, taking steps to explore the unknown and to find out the answers to his or her questions.[20]

These processes are frequently echoed in the descriptions of the ideal learners. For example, one respondent wrote that her ideal learner would "openly question society … why things are the way they are," and another stated that his ideal learner learns by "thinking critically about *everything*" (emphasis in the original). The best examples, however, are again found in the Western icons of scientists. From the Nobel Museum, we read that Leland H. Hartwell, who won the Nobel Prize in physiology/medicine in 2001 for his co-discoveries of key regulators of the cell cycle, recalled his burgeoning skepticism in his childhood: "I was an avid collector of bugs, butterflies, lizards, snakes, and spiders. I remember on one of these adventures learning to be skeptical of everything one reads in books. I had read that lizards do not have teeth. I had grabbed a very big lizard and stared in disbelief as it turned its head, displayed a fine set of teeth and sank them into my thumb."[21]

Similarly, Craig C. Mello, another recent Nobel laureate in physiology/medicine in 2006 for his co-discovery of RNA interference-gene silencing by double-stranded RNA, wrote on his critical thinking about a much larger topic:

> By the time I was in middle school, I had decided to reject religious dogma altogether. The "absolute knowledge" offered, was in my view, inadequate to explain the world around me. Furthermore, it seemed wrong to claim knowledge based on one's culture or upbringing. I saw the leap of faith involved in religion as smothering dialogue, closing the door on non-believers and walling them out of one's society. In contrast, the scientific method with its focus on asking questions and admitting no absolutes, was and continues to be refreshing to me. Science is grounded on, and values, dialogue. It is a human enterprise that breaks down walls and challenges its practitioners to admit ignorance and to question all ideas.[22]

It is important to point out that critical thinking in the West is not a value-neutral but a value-laden act. Unlike the more pure mental processes, critical thinking presumes a stance the person takes toward the world of knowledge and learning. This stance is also a direct manifestation of the Western notion of individual rights to question and to investigate anything, in any context. The only originating force is the questioner him- or herself having a desire to question. Critical thinking as learned and practiced in the West is only possible when such individual rights are granted and supported. As discussed in Chapter 2, Socrates, as a trying case, a pioneer in Western history, took as his life's purpose to question everything. During his own time, he was *not* celebrated by authority, although he amused many, particularly young people. His method of "eternal questioning,"[23] his great legacy, was retroactively admired and reified only by later generations. But because he was not guaranteed the right to do so by anyone, he was condemned to death for "poisoning youth" and for not believing in the gods of his city-state. Similarly, Western history is replete with persecutions of heretics (e.g., Bruno and Galileo) who dared to voice different interpretations and ideas of religious dogma.

Self-Expression and Communication

Self-expression and one's ability to communicate with others is another hallmark of the intellectual achievement of a learner in the West. All the active learning, exploration and inquiry, and critical thinking need to be expressed and communicated both as an integral part of the learning process and as the outcome of learning. The great emphasis in communication lies in the oral/verbal expression foremost, even more so than the written expression, although the latter is also of central importance.

There is a long tradition of oral eloquence that dates back to Greek antiquity through the Roman era and Christian tradition of delivering sermons to congregations (Chapter 8 is devoted to this topic). Modern democratic, academic, and business life all require the ability to speak publicly. Oral expressivity is not only taken as a sign of individual intelligence, but a skill that enables one to achieve just about any goal in life.[24] Accordingly, one needs to express one's thoughts and feelings in order to make friends, to date, to do well in school, to rise up the corporate ladder as a manager/leader, to articulate one's concerns in community living, to participate in democratic life, and to defend oneself in court or other official proceedings should one face such situations. Given the essential role of communication in the West, there is little wonder why learning how to self-express and to communicate is an essential task of any learner.

My lexicon study on European-American learning concepts yielded these terms of communication: *expression, communication, discussion, dialog, participation, interactive, listening, connection, engaging, collaborate, involving, illustrate, diagrams, essays, debate, and critique*. The descriptions of ideal learners images that I collected also showed that 70 percent of the respondents mentioned that these learners engaged in talking to others, discussion with professors and peers, and otherwise interacted with all kinds of people in all kinds of contexts (compared to only 30 percent of the Chinese respondents who referred to communication and discussion as a process of learning).

The socialization of self-expression begins early in Western learners' lives. Research on early mother–child communication indicates that European-American children, for example, talk more than their Chinese peers about their own qualities, thoughts, and feelings. European-American children's self-oriented talk is longer even when their total talk volume is less than that of Chinese children for the same controlled studies (in comparison, Chinese talk more about social relations and activities rather than themselves). When recounting past experiences, European-American mothers ask their children significantly more probing questions about how they lived through the experience, how they felt, what ideas they thought about, and how their individual selves were affected. As a result, as shown in research, these children express a great deal more of these aspects of their life experiences.[25] Within the family, a commonly documented opportunity to involve children in talking and discussion is dinnertime when family members sit together. Many children and parents attest that this is a very important social context in which parents and children engage in free exchange of their daily experiences. Children are encouraged to interrupt adults and to ask questions; children also chat with their siblings. Again, these repeated occasions serve to promote children's oral self-expressions.

Preschools in the West also make a great effort to develop children's speaking and expressive ability. Reggio Emilia schools as an exemplar, but Western preschools in general, very commonly encourage children to express themselves. When a child utters an idea, that idea will be recorded, listened and responded to, commented on, and followed up by a teacher in Reggio Emilia. Frequently a budding idea by the innocent child will evolve into a drawing, a sculpture, a poem, or a scientific project to engage the child.[26] This kind of attention to expressed ideas makes children value self-expression and motivate children to express more.

Across Western preschools, teachers frequently ask children how they feel, if they like a given activity, and what ideas and suggestions they have about a topic, which echoes home socialization as noted previously. The emphasis is on both the act of talking and on self-oriented talk. To achieve this goal, it is very common for teachers to tell children that there is no right or wrong answers and anything they say is fine. Since the notion of emotional intelligence was introduced in the 1990s,[27] Western preschools endeavor to help children articulate their emotions, particularly negative emotions, into words rather than acting on such emotions. For example, if a child feels angry about a toy that he or she cannot get from a peer, the child is encouraged to say what bothers him or her and how the situation makes him or her feel, instead of taking direct action such as fighting, grabbing, kicking, or punching the peer in order to achieve the goal of getting the toy. When a child has committed a transgression, the child is likely to be asked to apologize *verbally* to the teacher or peer. Similarly, when a child receives a nice act from another child or an adult, the child is also highly encouraged to *say* "thank you" to the other (rather than to return a kind *act* to the other such as giving the peer a toy or candy – the latter is a more common nonverbal expression of gratitude in Asian socialization).

In grade school, while continuing their learning in verbal self-expression, children begin learning how to express themselves in writing. As soon as they learn how to use words and make sentences, they are encouraged to write about their qualities, thoughts, relationships, and feelings. They are asked to write their own stories, describe their own families, and present their own ideas. When they advance into middle and high school, children are taught writing skills for argumentation. One important format for developing such skills is essay writing, where children are taught to come up with a thesis to support, gather evidence, present counterarguments, and substantiate the merit of their thesis.

In middle school, and even more in high school, children also learn how to debate. In class, one teaching method is to divide the class into groups, each of which takes a different side of the topic/issue being taught. After the groups have mustered counterarguments against the other group's position or issues, the class meets again to start an open debate. Not only do students enjoy this form of oral argumentation, but they also learn an essential skill. In all public and private schools, there are debate teams for children to join. Children are also taken to tour through the towns and states in order to experience different audiences. Local, state, and national competitions are held. The winners are much admired and celebrated.

In higher education, particularly in the liberal arts, students engage in formal training in public speaking. Every university offers such courses, with many as requirements for graduation. Where no such requirement exists, students stand in line to enroll, attesting to the unabating popularity of such training. Such courses provide students with the opportunity to analyze great orators in history, such as Martin Luther King, and to write their own speeches to deliver to the class. Teachers and classmates critique each speech to help students improve their craft of public speaking. Other than such formal courses, most liberal arts courses require that students give oral presentations of their projects in addition to writing papers. Finally, Western learners are also encouraged to use all visual forms such as diagrams, illustration, videos, and sound to enhance their presentational impact.

These four key components of the Western learning process are also resonated in the descriptions of the ideal learners in my study. Taking these elements together, European-American respondents almost unanimously (96 percent) made references to such mind-oriented learning processes. In comparison, only 68 percent of their Chinese peers did so.

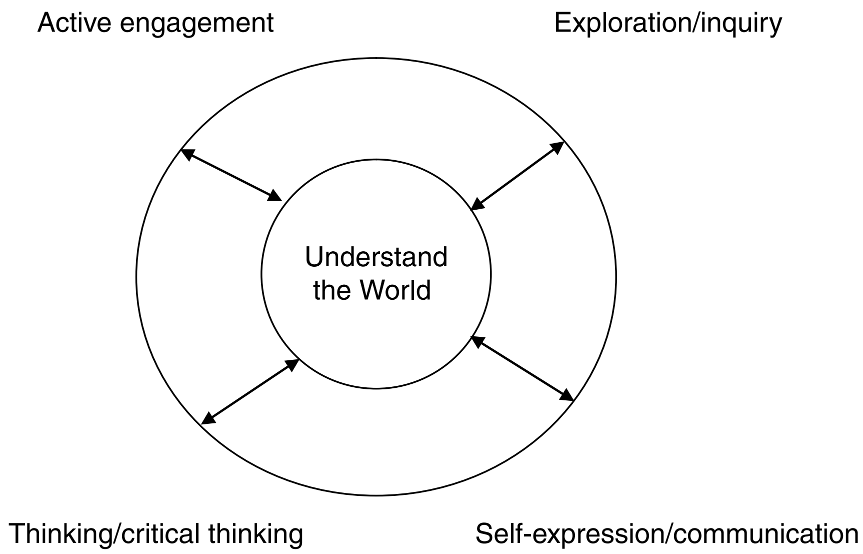

FIGURE 4.1. Diagram for mind-oriented learning processes in the West.

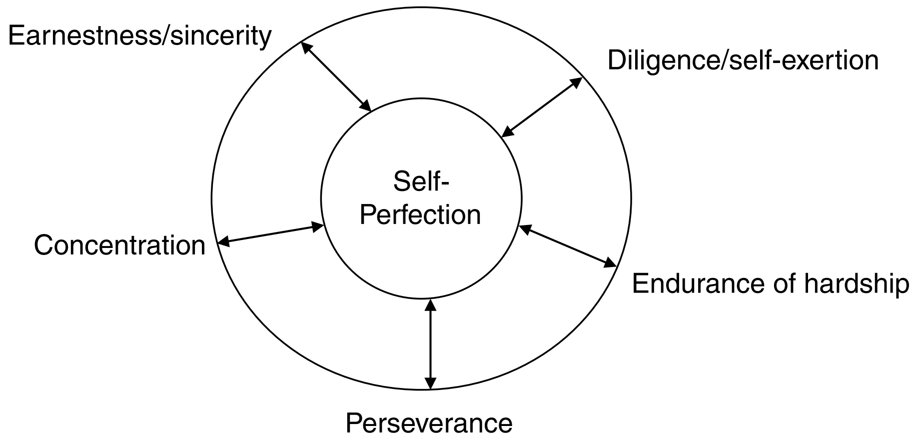

FIGURE 4.2. Diagram for virtue-oriented learning processes in China/East Asia.

Virtue-Oriented Learning Process in China/East Asia

In describing the virtue-oriented learning process in China/East Asia, I focus on the five key virtues of earnestness/sincerity, diligence/self-exertion, endurance of hardship, perseverance, and concentration (see Figure 4.2. Respect and humility are treated in Chapter 5).

Earnestness/Sincerity (認真/誠意)

As noted in Chapters 2 and 3, sincerity has been regarded as the first virtue in order to dedicate oneself to learning in the Chinese tradition. Sincerity was already clearly expressed in the *Analects* when Confucius discussed learning with his students. In *Mencius* and the writings by Xun Zi, sincerity was emphasized, but it was more centrally and clearly articulated as a required virtue for self-cultivation by the Neo-Confucians. Earnestness is the more modern term used to describe this initial commitment to learning that can also be applied to young children (sincerity as a term would be too abstract for them). Earnestness/sincerity means to take one's learning to heart and to be serious about learning. In other words, this virtue aims at making one instructable, teachable, receptive, and open to learning while putting one's ego out of the way.

How, then, do Chinese/East Asian learners demonstrate earnestness/sincerity in their learning? My learning lexicon study[28] produced the following six terms/expressions from daily usage that give a first account of the process: *earnestness (認真)*, *learn earnestly and make daily progress (好好學習, 天天向上)*, *one shall not fear one's ability to learn; one shall fear one's lack of desire to learn (不怕學不會, 只怕不去學)*, *a cause is perfected with diligence and neglected by frivolity (業精於勤, 荒於嬉)*, *effort never lets down the resolute person (功夫不負有心人)*, and *there is no difficulty on earth for the resolute person (世上無難事, 只怕有心人)*. What these expressions have in common is their emphasis on earnest and sincere dedication as contrasted to lack of such (e.g., frivolity, neglect, and lack of resolution).

From observational research on Chinese/East Asian classrooms, it is clear that the learner, even a very young child in preschool, needs to cooperate with the teacher in the classroom. If asked to sit down, to listen, to look at the board, to take notes, to respond, and to do homework, the learner should do so without resisting such instructions. Learners need to give the instructions and learning activities their full attention. In this kind of cooperation, learners should not interrupt/disturb teaching and other children's learning, such as chatting with peers, laughing, moving out of a seat, or leaving the classroom while the teacher is teaching. Students' earnestness/sincerity is needed not merely for their own learning, but also for respecting other children's learning.[29]

In such a setting, learning is taken not just seriously but also sacredly. It is not exaggerating to say that this seriousness is analogous to other children's "right to learn" and is imbued with all the moral meanings noted in Chapter 2. This "right" cannot be violated or taken away by anyone, even though there is no law to enforce this "right." But for the student inclined to disturb, it is his or her obligation to respect that right in others. Likewise, his or her own learning is also regarded as sacred and inviolable by other children. But because it is part of a strong cultural norm, there is no need to have written law for its protection. To use another analogy, this kind of commitment and seriousness in regard to learning is also similar to a

Western orchestra being conducted by the conductor who demands and actually receives the full cooperation and earnestness from the players. The mutually understood roles and responsibilities/obligations between the conductor and the players are guaranteed by everyone to such an extent that the whole orchestra would collapse if even one player no longer wanted to honor this cooperation. This resemblance probably made American education researcher Lynn Paine call Chinese classroom teachers virtuosos of performance. The kind of teaching described in her research was possible only when the learners exercised their virtue of earnestness/sincerity. Understood as such, there is no wonder why disturbing teaching and other children's learning by any student is regarded as a serious learning problem in East Asia.[30]

It is erroneous to assume that Chinese/Asian children must not enjoy such learning or that they must suffer from lack of freedom or creativity. Research indicates that they derive happiness and satisfaction and show enthusiasm even when they have to demonstrate their mistakes on the board and respond in chorus (a uniform learning activity instead of individualist activities).[31] Their joy and satisfaction are probably similar to the emotions of orchestra players when they produce a great performance under the direction of a conductor.

If homework is assigned and the student fails to do it for no legitimate reason, that behavior is regarded as lack of earnestness/sincerity. Further, even when the student does the homework but rushes through it, showing impatience (e.g., illegible handwriting) and carelessness, that behavior is also regarded as lack of earnestness/sincerity. Therefore, any learning behavior that is just going through the motions without the dedication of the heart, the affectively charged receptivity to learning, is interpreted as the learner not having earnestness/sincerity and consequently not being ready to learn.

An earnest/sincere learner does not stop at just following these basic expectations; he or she goes a step further. Ideally, he or she also takes initiative and seeks out ways to exceed such expectations. For example, such a learner may make a great effort to take good notes. To learn the material, he or she may visit a peer to ensure that his or her original notes are complete. He or she may seek out the teacher to clarify unclear concepts or resolve confusion. When learning that a peer may have deeper or clearer insights into the material, the learner may make an extra effort to learn from the peer. Whenever a learner endeavors to make sure that he or she acquires the skill or material in question and whenever the learner actually takes the various steps needed to achieve the goal, he or she is believed to show earnestness/sincerity in learning. As recounted in Chapter 2, the two students traveling from south to north across China to seek out Master Cheng's teaching and forced to stand in the snow during the Song time (程門立雪) were taken as a good example of displaying sincerity and dedication to learning (in addition to respect for the teacher). Thus, a great deal is expected of the learner in the ways of taking learning seriously as an initial engagement in learning anything from anyone.

Diligence/Self-Exertion (勤奮/發奮)

There are several terms in Chinese (努力/勤奮/用功) that are used very commonly and interchangeably to refer to the notion of hard work or effort (with the first term also commonly used in Japanese and Korean). Recall that the nearly 500 English learning terms I collected in my lexicon study did not contain either *hard work* or *effort*, but the Chinese lexicon contained 46 such terms that entered the core list (many more such terms were not included because of lower group ratings). The phenomenon of showing elaborations of a given conceptual domain in one culture but not in another was conceptualized as *hypercognition* versus *hypocognition* by Robert Levy who studied emotion terms in Tahiti.[32] Clearly, the conceptual domain of hard work/effort is hypercognized in Chinese culture (and other East Asian cultures) and hypocognized in the West (at least as captured and nominated in the learning terms of English). However, English *hard work/effort* hardly do justice to the many terms that refer to finely differentiated and detailed virtue-oriented learning processes of Chinese learners. To highlight this difference, I use the terms *diligence/self-exertion* instead of *hard work/effort* to discuss the various learning processes involving this virtue.

Upon careful inspection, the virtue of diligence/self-exertion describes at least three learning processes leading to three connected goals: (1) familiarizing oneself with the learning material (*shu*, 熟), (2) practicing (*lianxi*, 練習), and (3) achieving refined/perfected mastery (*jing*, 精). In the following subsections, I detail each of these processes.

Familiarizing Oneself (shu, 熟)

Familiarizing oneself is the first step to take whenever Chinese learners encounter new material, a new concept, or a new skill. Familiarizing oneself with the new material is necessary to ensure that the learner recognizes all the pieces, components, and details of a given task. For example, if a learner is introduced to a new song, a new poem, a new list of English words, or a new math proof, he or she usually begins to look at/read/sing each component. Next, the learner may begin chanting the poem, reading aloud each English word, or writing each math step to him- or herself, and he or she is likely to do so over and over again. Thus, we have the terms from my lexicon study: *one's mouth won't be fluent without chanting for three days; one's hand gets out of practice without writing for three days* (*三天不念口生, 三天不寫手生*), *well-versed in three hundred Tang poems, one will be able to chant if not compose poems* (*熟讀唐詩三百首, 不會做詩也會吟*), *familiarity leads to skillfulness* (*熟能生巧*). Familiarity *shu* is juxtaposed with its opposite state *sheng (生/生疏)*, meaning rusted, unfamiliar, or out of practice.

The process of familiarizing oneself is believed to be the necessary first step of learning anything because it ensures fluency and first-level knowledge of the material. If one is to make any progress upon encountering the new material, the measure of it would be a step beyond unfamiliarity. Otherwise, one might as well not have encountered the new material at all. The idea of familiarity is not foreign to Western learners. It is the same idea as expressed in the common English phrase *having it at my fingertips*. The purpose is to make oneself familiar with all the "surface" details of a given learning task, not necessarily – or often even deliberately not – to aim for any deeper understanding of the material. The end result of familiarizing oneself is often a measurable achievement: memorization, or learning the material by heart.

Practicing (lianxi, 練習)

Practice is the absolutely central behavioral manifestation of the learning virtue of diligence/self-exertion in Chinese/East Asian learning. Based on what I can gather from available research, the notion of practice has four further processes aiming at four respective goals: (1) ensuring the length of time for practice that is perceived as necessary to accomplish the goal (e.g., the ten years in "ten years off the stage practice," *台下十年功*), (2) reaching the desired cumulating achievement (e.g., the one minute in "one minute on stage performance," *台上一分鐘*), (3) reviewing old material to gain new insight (*wengu zhixin*, *溫故知新*), and (4) offsetting one's lack of talent and ability (*buzhuo*, *補拙*).

Ensuring the required length of time for practice. After the learner has achieved some familiarity with the new material, he or she is likely to submit him- or herself to the rather prolonged and often arduous process of practice.[33] Again, my lexicon study revealed a glimpse into the nature of this part of practice: *lay down effort* (*下功夫*); *one minute on stage performance, but ten years off stage practice* (*台上一分鐘, 台下十年功*); *ten years of cold windows (studying)* (*十年寒窗*); *always have a book in one's hand* (*手不釋卷*); *never stop punching (practicing martial arts) and never stop singing (practicing opera)* (*拳不離手, 曲不離口*); *if you work at it hard enough, you can grind an iron rod into a needle* (*只要功夫深, 鐵杵磨成針*);

without accumulating small steps, one cannot reach a thousand miles/without gathering little streams, there cannot be rivers and seas (*不積跬步無以致千裡, 不積小流無以成江海*); *it takes more than one cold day for the river to freeze three feet deep* (*冰凍三尺, 非一日之寒*); *long-term diligence is the road to the mount of knowledge* (*書山有路勤為經*); *effort never lets down the resolute person* (*有心人*); *do one's utmost to self-study* (*勤勉自學*); and *Wang Xizhi practiced calligraphy (his clothes were damaged by his fingers practicing on them) (王羲之練書法).* All of these expressions highlight the key meaning of practice: The learner must recognize the importance of the long, dedicated process required for learning anything to which one has given earnestness/sincerity. Learning is not just a serious matter; it is a *long* endeavor. There are no shortcuts in learning.

Some of these expressions came from Chinese intellectuals such as Xun Zi and exemplary cultural icons of talent and learning such as Wang Xizhi. Most of these expressions reference ancient skill and artistic learning such as painting, calligraphy, martial art, dancing, and opera. Admittedly, such skill learning requires long-term practice regardless of time and space. A musician learning how to play the piano now would need to practice just as much as someone 200 years ago if he or she wants to achieve any level of mastery. There is also a shared view that any worthwhile knowledge/skill needs a solid foundation without which deep understanding is not likely to occur. Furthermore, such a foundation cannot be established overnight, but requires a long time. The most commonly recognized approach to such a foundation is long-term practice.[34] However, the impact of these expressions does not lie in actual skill or knowledge being learned, but a general adherence to diligence/self-exertion of practice as a *virtue*. As such, the learner needs to truly engage in the actual process of practice.

Keeping this long-term process in mind, Chinese learners work at each component, each key step repeatedly, over and over again, until they feel that they have achieved desired understanding or mastery. Then they move to the next component or step where they are likely to repeat the cycle of practice. Although there is much such recycled practice, it is not approached mindlessly, but carefully and attentively. Attentiveness and mindfulness are encouraged and admired.[35] With each practice, the learner needs to observe their own progress, although progress may be incremental depending on the actual skill or knowledge under study. Tolerance toward incremental progress is also important. A "great leap forward" in achievement without dedicated work is not believed real. Likewise, looking for tricks or shortcuts to circumvent this long practice is taken as a sign of lacking diligence/self-exertion. Moreover, the previously exercised earnestness/sincerity is not only expected to continue, but also to intensify. Because practice requires much more effort, earnestness/sincerity is assumed as an integral part of diligence/self-exertion.

Indeed, research on Chinese/East Asian learners in learning modern school subjects such as math and science documents that learners not only view the learning process as requiring effort over a long time, but they also engage in prolonged and repeated practice (with math and other science problems). For example, the well-researched topic of time spent on learning provides ample evidence that Asian learners spend much more time doing homework, reviewing covered material, and preparing for new material and assessment than their Western peers. They also spend much more time on practicing skills and materials outside class, that is, they practice above and beyond completion of assigned homework.[36]

Reaching the desired culminating achievement. All that practice is geared toward one goal: achieving understanding and mastery of the desired skill or knowledge under study. As the research by Dahlin and Watkins shows, Chinese learners, unlike their Western peers, do not believe that deep understanding or mastery comes from a sudden "ah-ha" moment,[37] although Chinese learners also experience such moments, as the term s*uddenly see the light after being stuck* (*茅塞頓開*) indicates (which also entered my lexicon study). By looking at the terms in the preceding section, it seems that Chinese learners engage in long practice in order to get very deep and solid in their understanding and mastery. Consequently, one aims at reaching the condensed *one minute on stage* out of the *ten years of practice*, all accumulated knowledge from the *ten years of cold windows*, the *deeply frozen water* from many days, and the *thousand miles* from all the *small steps* as well as the *river/sea* from all the *streams of water*. Apparently, it is the breadth and depth of this result that they desire, not any small understanding or mastery. And breadth with depth is indeed a key goal for achievement in the Chinese learning conceptual map (see Figure 3.2).

But the question remains as to whether Chinese learners can really achieve this type of deep understanding and mastery. After all, the stated goals are rather vague; they seem more metaphorical and aesthetically worded than any real benchmark for achievement. If these goals are not obtainable, then why do Chinese learners still engage in this type of diligence/self-exertion? Two responses to this question can be given. First, it is quite possible that such learning goals can indeed be achieved, even though they may remain vague in the mind of the learner. Take the well-known case of practicing martial arts as an example. Even though one may not know for sure if one *will* achieve a high level of the skill, one still has a clear idea of what that end goal looks like, because the teacher embodies the skill. Moreover, one can see high levels of mastery in other highly accomplished martial artists. Thus, so long as the learner continues to practice, especially when he or she has been told that such practice was how the teacher and others achieved their mastery, that learner may indeed see that he or she can achieve the goal eventually. This is very similar to Western learners desiring to learn golf, gymnastics, figure skating, ballet, classical music, and instruments. Such learners exert effort not necessarily knowing that they *will* achieve a given goal. But they do know what to aim for because they know the accomplished musicians and athletes. They, too, are likely to engage in this kind of long, arduous practice if they are told that this kind of practice is needed for achieving anything in these domains. Learning other subjects such as math and science may also lead the learners to engage in long practice, because Chinese children are always given the causal link between long and hard practice and later achievement (whether such causality is true is beside the point). They receive this kind of deliberate socialization regardless of the discipline and regardless of how much talent a domain of work requires (see Chapter 7 for a focused discussion). Even great scientific discoveries are introduced to children as possible only when the scientists know the details of the topic and are deeply engrossed in their daily work.[38]

Second, in keeping with the Confucian learning ethos, the learner may not care much about the end goal because it is vague, and perhaps kept deliberately so (one wonders why Chinese have not made their learning goals measurably explicit for millennia). In all likelihood, learners may be quite clear that they *will not* achieve a high level of mastery just as they know that not all students will get an A in school. But as research shows, Chinese learners are still very willing and quite possibly happy to engage in diligent practice. For one thing, they know that they cannot get worse; they can only get better with practice. For the other – and most importantly – displaying learning virtues, as I have argued elsewhere, may be a highly motivating goal by itself regardless of whether the end goal of mastery will be reached. A learner's behavior is not judged retroactively positive or negative after his or her achievement has been determined, but as he or she engages in learning from the very beginning. Outcome cannot play a role because there is no outcome yet. Thus, showing earnestness/sincerity and diligence/self-exertion alone is inherent goodness and therefore reason for social reward and admiration. This indeed happens in Chinese schools. I personally witnessed the following scenario of honoring virtues despite low achievement:

> In middle school, my class took a math test. I had one of the highest scores in my 60-student class. The teacher told the class that he wanted to honor a great student for this exam. I thought it was me whom he was going to honor. But instead, he asked a quiet girl sitting in a corner to stand up. Everyone knew that she was not the highest achieving student in class. But the teacher said "Class, let's give Yulin a round of applause for her great diligence for this exam. Although she did not do very well, her spirit is honorable. We all can learn from her."

The message from the teacher was very clear: It is not the outcome of one's achievement that calls for admiration, but one's diligence/self-exertion. The fact that one has given one's complete effort even without achieving stellar results in the process is cause for respect. In fact, high achievement or high intelligence without virtues is not admired very much. When asked to respond how their ideal learners would learn if they achieve highly or possess high intelligence, Chinese respondents wrote that they will continue to work hard to aim at even higher goals. High achievers would not feel full of themselves because their goal is to self-perfect for life.[39]

Ngai-Ying Wong provided a description of the Chinese approach to practice that further explains the point of practicing for Chinese learners. Accordingly, Chinese learners first try to "enter the way" (入法). For example, when learning calligraphy, a learner hopes to get a feel of the skill, that is, how to apply ink, force, attention, movement of one's arms, and so forth to produce similar work to that of the master that the learner admires (this is very similar to, say, Western golf learners trying to imitate a master). One will not stop one's practice until one can produce the desired result. However, this imitative perfection is only half of learning, not the end goal. Once satisfied with one's practice, one moves on to a higher level called "existing the way" (出法), retaining but transcending the imitative achievement.[40] A learner who has gone through both processes is said not only to have mastered the skill, but also to have developed a personal style. The latter is similar to Western classical painting. A novice painter, too, had to practice for a long time in order to imitate the master; it was only through rigorous practice that the painter could go beyond the original style to create something new. But this new style is also deeply grounded in the original master's techniques and skill. Thus, it seems that Chinese learners are quite methodical in their learning with the virtue of diligence/self-exertion.

Reviewing old material to gain new insight. Chinese learners inherit a strong principle, as taught by Confucius himself, that learning is a cumulative process. Anything one has learned in the past is actually not gone factually, but the learner may not benefit from that past learning experience if he/she does not reflect on it anymore. However, just thinking or reflecting on it may not lead to the best learning either because reflection is a pure mental exercise. Given that Confucian learning strongly emphasizes moral learning and practice, pure mental exercise is not regarded as the best learning; the learner needs to go a step further, reviewing the old to gain new insight. Thus, "reviewing the old as a means of realizing the new" as stated by Confucius is regarded as an important learning principle that virtually all Chinese children are taught to say and to use for their own learning.

This principle focuses on two related aspects. First is the review part. Accordingly, it is one's responsibility to ensure mastery and related further learning of whatever one has been taught or has learned. Review of learned material enhances familiarity, memory, understanding of the material itself, seeing the need to practice, and other related elements. By reviewing, one also engages in reflection, checking one's understanding, making connections, and synthesizing. The process of review necessarily leads to deepened understanding, hence the notion of "new insight." New insight as a concept cannot be anything else but deeper understanding. In fact, new insight is the real purpose of learning by review. Thus, from the very beginning, Chinese learning has been to aim at understanding and insight. The difference may lie in their focus. Unlike their Western counterparts who are likely to delve into mental understanding, Chinese learners focus on moral insight into self-perfection and other related learning. Confucius himself regarded this learning-by-review as such an important process that he considered those who could gain new insight by reviewing as those who could be or become teachers (溫故而知新, 可以為師矣).[41]

Review is thus a part of practice with the virtue of diligence/self-exertion. Furthermore, it should not be surprising that Chinese learners assume autonomy and personal agency in achieving this very high goal of learning. Teachers and parents openly emphasize review and provide opportunities for children to do so. In every Mainland Chinese school (and in most schools in other Chinese regions), there are scheduled self-study classes (自習課) every day, where students sit quietly in class, doing homework and reviewing what they have learned. Outside school, students also engage in review on their own. Research has consistently found that Asian students spend more time studying outside class and school than their Western counterparts. A recent study of Vietnamese and German students' learning activities also showed that Vietnamese students spent much more time reviewing material they learned in class than their German peers.[42]

Offsetting one's lack of talent and ability. In their informative research on why Asian students achieve better than American students, Stevenson and Stigler[43] found two key concepts. Whereas Western children believe in inborn ability, Asian children believe in effort. Research on Japanese education also indicates that they downplay inborn ability and emphasize effort.[44] The questions that I have heard people in the West ask often are: Do Asian people not know that individuals are actually born with different levels of intelligence? Or do they not care? If so, why not? These questions were indeed left unanswered in previous research.

As it turned out, Chinese learners are keenly aware of such individual differences, and they also know that these differences may give people advantages in learning. No one doubts that one needs talent to be a great athlete, artist, poet, writer, musician, or scientist. Few also deny that there are prodigies and wiz kids, and that some people have better memory, faster mental processing speed, more energy, ability to achieve higher work efficiency, and other superior abilities. It is also not the case that Chinese learners do not care about these important differences. They do care, particularly with regard to their need to know where they stand in their learning progress when they are grouped with high-talent individuals. However, they do not believe that just because they are not prodigies or lack apparent talent, they cannot learn much. In fact, they believe that they can learn very much if they apply diligence/self-exertion. They do not aim at becoming the next Einstein, but they can still achieve much scientific learning if they work hard. If, however, they do not work hard, then they can surely bear the consequence of not knowing much.[45]

In sum, Chinese learners are not oblivious to individual differences, and they actually care about these differences. They do not, however, care about them to the point where it has a debilitating effect on their process of learning. They do not lament their own lack of inborn ability, and they disagree that only talent can lead to great achievement. From childhood on, Chinese learners are exposed to many exemplars throughout history who have achieved great learning with personal virtues. Many of these exemplars had disabilities such as blindness, deafness, and other physical handicaps. These role models are contrasted with people who have "normal" intelligence, or even superior talent, but have not studied hard. What distinguishes the two kinds of people in their learning achievement is the level of exercising their learning virtues.[46]

Thus the fourth goal of practice is to offset one's perceived lack of talent or ability. There were two phrases in my studies that refer to this purpose of practice: *Diligence can offset one's clumsiness* (*勤能補拙*) and *the* *clumsy bird flies early* (笨鳥先飛). These phrases are told to children again and again, along with examples of real learners. If a person lacks talent or sufficient ability, that person will increase diligence. In my study on ideal learners, I asked the respondents to describe how their ideal learners would learn if they were not very bright naturally. Their descriptions indicated that their ideal learners would double or triple – and in some cases increase a hundredfold – their diligence (百倍努力). The respondents unambiguously expressed respect and admiration for those who perceived lacking ability or talent and decided to increase their diligence in order to achieve their learning goals.[47]

Achieving Refined/Perfected Mastery (jing, 精)

All that familiarizing and practice are not even the end goal. There is the ultimate goal toward which the first two processes are directed: refined and perfected mastery (*jing*, 精). Thus we have a set of Chinese expressions about this ultimate height of achievement that entered my lexicon core list. The first is *familiarity leads to skillfulness* (熟能生巧) *and skillfulness leads to eventual perfection* (巧能生精). While the first part has been written about quite a bit, the second part of the expression has not been discussed much. Refined/perfected mastery, *jing*, is a state that does not lend itself to clear definition or articulation because it is a state only the greatest masters can reach and display. This kind of virtuosity is found in every domain of human pursuit whether it is in Western, Asian, or any other culture. Other expressions show various aspects of this type of refined and perfected learning: *thorough understanding* (透徹理解); *refined understanding* (精通); *know something inside out* (滾瓜爛熟); *profundity* (精深); *striving for perfection* (精益求精); *superb achievement* (出神入化); and *reach the highest degree of refinement* (爐火純青). The goal is not just any understanding, but lucid and thorough understanding; not just mastery of the skill, but the refined/perfected skill; not just knowing something, but knowing it profoundly.[48]

However, the highest *jing* is knowing something inside out and using it *effortlessly* and at the same time perfectly. This way of working and performing has become the second nature of the person; it is the merging with the Way in both Confucian and Taoist persuasion. The best illustration is the story *Chef Ting Carves Bulls* (庖丁解牛) by Chung Tzu (369–286 BCE), Taoist philosopher and a contemporary of Mencius:

> King Hui of Wei had a chef named Ting. When Chef Ting was carving a bull for the king, every touch of the hand, every inclination of the shoulder, every step he trod, every pressure of the knee, while swiftly

> and lightly he wielded his carving-knife, was as carefully timed as the movements of a dancer.... "Wonderful," said the king. "I could never have believed that the art of carving could reach such a point as this." "I am a lover of Tao," replied Ting, putting away his knife, "and have succeeded in applying it to the art of carving. When I first begin to carve I fixed my gaze on the animal in front of me. After three years I no longer saw it as a whole bull, but as a thing already divided into parts. Nowadays I no longer see it with the eye; I merely apprehend it with the soul. My sense-organs are in abeyance, but my soul still works. Unerringly my knife follows the natural markings, slips into the natural cleavages, finds its way into the natural cavities. And so by conforming my work to the structure with which I am dealing, I have arrived at a point at which my knife never touches even the smallest ligament or tendon, let alone the main gristle.
>
> A good carver changes his knife once a year; by which time the blade is dented. An ordinary carver changes it once a month; by which time it is broken. I have used my present knife for nineteen years, and during that time have carved several thousand bulls. But the blade still looks as though it had just come out of the mould. Where part meets part there is always space, and a knife-blade has no thickness. Insert an instrument that has no thickness into a structure that is amply spaced, and surely it cannot fail to have plenty of room. That is why I can use a blade for nineteen years, and yet it still looks as though it were fresh from the forger's mould.
>
> However, one has only to look at an ordinary carver to see what a difficult business he finds it. One sees how nervous he is while making his preparations, how long he looks, how slowly he moves. Then after some small, niggling strokes of the knife, when he has done no more than detach a few stray fragments from the whole, and even that by dint of continually twisting and turning like a worm burrowing through the earth, he stands back, with his knife in his hand, helplessly gazing this way and that, and after hovering for a long time finally curses a perfectly good knife."[49]

The phrase "the knife can surely find plenty of room" (游刃有余, i.e., accomplishing a task with ease), still used widely in daily Chinese, comes from this story. As can be seen, this level of knowing something and using it freely and effortlessly indeed requires much practice, much mindfulness, and much dedication. There is little chance that blind practice without engaging in careful study, attention to details, contemplation, and self-exertion would ever reach the level of *jing*. Moreover, chances are that this carver derived no less personal and intrinsic enjoyment and satisfaction, as well as admiration and acclaim, from his social world than did our contemporary masters such as Yo-yo Ma, Ben Kingsley, or Michael Jordan.

Having presented the intricate learning processes under the virtue of diligence/self-exertion, I wish to address the oft-condemned Asian rote learning. When Westerners visit Chinese schools and observe how Chinese/Asian children learn, they can clearly see the familiarizing and practicing process: Behaviorally, it is nothing more than repeating, reciting, and copying things – that is, acts of committing the material to memory.[50] On the surface, such learning behaviors do not show any breakthrough in thinking and discovery. If the observer does not know that there is still a very long list of learning virtues and steps to follow, he or she is likely to conclude that Chinese learning is all rote learning. Together with the also observable teacher-dominated classroom and parent-dominated home learning environment, it is even easier to feel sad and indignant for Chinese/Asian children. There are no apparent freedom, creativity, exploration, inquiry, and self-expression – that is, forms that indicate personal agency in learning. Chinese/Asian learning appears indeed pathetic.

However, there are some major flaws in the reliance on such observations (both empirical and passing impressions) alone. If what is observable is taken as mirroring what is going on inside the learner, then these observers commit the same error for which behaviorism has been criticized. Interpreting what is observed without checking with the learners' own experience, aspirations, and interpretations is a highly subjective act that is colored by one's own cultural and personal lens.[51] Here we have the perennial problem of substituting etic (outsider's) for emic (insider's) meanings. When I first encountered Western interpretations of Chinese/Asian learning, the prevalence of this problem was overwhelming. There was a complete disconnect between these observations and interpretations and my own experience of learning in China. Yet, such research has persisted to date.

The second fallacy in condemning Chinese learning is that researchers forget that much of valued Western learning is also similar to Chinese learning, and such learning is foundational because it is necessary. As alluded to earlier, Western learners of piano (and other instruments), ballet, sports, phonics, many math problems for a given concept, introductory biology learning, and the notorious medical school beginner courses do just as much of this "blind" and "meaningless" rote learning and repetitive acts. There is likewise just as little freedom, exploration, inquiry, discovery, and creativity. And there just does not seem to be any way to avoid such learning or to transform it into more creative and exploratory learning. Are we to claim that these learners are also stripped of their personal agency, or that they too must suffer from their teachers' authoritarianism? It is peculiar that once a genius arises, even after much of this type of foundational learning took place, the hard work of geniuses is usually edited out of the discourse of the accolades in the West (but not in East Asia). Sergei Bertensson, Jay Leyda, and Sophia Satina provide a telling biography on Rachmaninoff's early training for piano under Nicholai Zverev:

> Rachmaninoff and his peers, Leonid Maximov and Matvei Pressman, had to get up at 6 to practice piano for several hours. The schedule was not allowed to change even when a boy had gone to bed at 2:00 am the night before. These boys took turns facing the ordeal for the one piano in the house. When a note was mishit by any boy, the teacher would storm out shouting. And when he was out of town, his strict wife continued the drill.[52]

As illustrated by this story, hard practice is often a necessity at the beginning. The longer a given discipline such as philosophy, history, math, biology, or medicine has been in existence, the greater the accumulated body of knowledge that the discipline has, and, consequently, the more foundational learning is required. Learners eventually reach more sophisticated levels of mastery and excellence if they continue the so-called rote learning over time.

The third fallacy that such observers commit is to assume that what one observes does not change. When a Chinese learner is at the familiarizing stage, the learner is likely to show acts replete with repetition and memorization. Several days later, however, this learner may be able to explain what a given poem means and how the words and sounds are put together, which is a level of understanding and insight. In another few weeks, the same learner may have learned several different forms of poetry and may be able to compare and contrast the variations. Finally, after the whole semester, this learner may be able to compose his or her own poems. Whereas some early trials may resemble the masters' styles, later trials may be quite unique and creative. None of this later learning is the same as the first phrase of familiarizing and practice. Yet, the observer takes the one observed incidence in the beginning and falsely assumes that the learner is a perpetual repeater and memorization robot for the rest of his or her learning, not just for the given skill but for all learning throughout the learner's life.

The fourth fallacy that such observers commit is to assume that understanding must be immediate, either achieved immediately during the learning activity or displayed verbally or behaviorally, or both, by the learner. As stated earlier, understanding can have many stages and facets, some of which may even require cycles of thinking, reflection, practice, application, thinking again, revising the original understanding,[53] and so forth, which is the essence of reviewing as a means to gain new insight. Understanding depends on the nature of the knowledge, concept, or skill under study. I would argue that the understanding that is observable within a class or a given activity in a limited time (e.g., a class period of fifty minutes) is a low level of understanding that is quite possibly not very worth pursuing because it does not require deep thinking and dedication over time. The kind of understanding that Chef Ting displayed cannot be achieved and therefore observed within a short period of time. Rushing to label and condemn Chinese/Asian learners as lacking understanding during a stage when their teachers or they themselves do not even expect to achieve any worthy level of understanding is simply invalid.

Some researchers have tried to distinguish memorization for the sake of memorization from repetitive learning. Whereas the first kind is truly rote learning, the second type can be meaningful and can lead to understanding and skill refinement. These researchers further demonstrate that Chinese learners use memorization for creating a deep impression as their first step. Then they seek deeper understanding of the material. Ference Marton's team pioneered this line of research with phenomenological methods and collected learners' own thoughts and experiences. They found that Chinese learners indeed engaged in memorization, but memorization was not an end in itself, but rather the first step. With information committed to memory, they sought to understand the material. One student in Hong Kong attested to such a process: "[B]ecause each time I repeat, I would have some new idea of understanding, that is to say I can understand better."[54] Another recent study comparing Vietnamese and German students shows that repetitive learning is strongly correlated with deep learning strategies for Vietnamese students, but hardly so for German students. The researchers concluded that repetitive learning is not a mechanical and thoughtless activity for Asian students.[55]

It was a relief that researchers actually found evidence that Chinese/Asian learners were not just mindless learners, but actually were inclined to and did in fact understand what they studied. Of course, the need to find such evidence only came from the apparent paradox of the independent and repeated demonstration of Asian children's achievement in international assessments. In all of this effort, the worry that perhaps we would not find such deep learning that would vindicate the Chinese/Asian learners' methods also showed the power of the fallacious but unfortunately widely accepted condemnation that has been directed at Chinese/Asian learning for so long. The irony is that the learners themselves, teachers, and parents do not seem bothered by it. Why should they? After all, who would not admire Chef Ting in Chuang Tzu's story? And who would remain untouched by the young Rachmaninoff and his peers when they practiced piano in cold Russian winters with such discipline?

The virtue of diligence/self-exertion is the utmost learning process in Chinese learning that affords the learner a chance to be like Chef Ting and other masters. It is also clear that this process aims at a very long time-span for a very high level of understanding and mastery. Such a time-span and aim obviate passing observations of a class period or a child murmuring a poem. Practice understood as such may appear rote and boring, but to the learners it may be uplifting and gratifying.

Endurance of Hardship (刻苦)

The idea that learning is fun or should be a fun process can be difficult for Chinese/Asian learners to absorb.[56] Based on the research I have reviewed and done myself, very few Chinese talk about learning in this way, even after so many Western children's TV and online fun-orientated programs have been running in China for several decades (longer in Taiwan, Hong Kong, and Singapore). Chinese may accept some fun for young children's learning, but probably not for older children. Why would they not feel that learning should be fun? In my opinion, this stems from two related convictions. First, as discussed in the preceding chapters, learning is not just academic for Chinese/Asians; it is, more centrally, a personal moral obligation and commitment. It is a very serious endeavor. It would be odd to say that striving to be moral, to become a better person, or to take on the obligation for life is fun. Given this ultimate purpose, Chinese, in my view, rightfully regard learning not as a lighthearted process but as a weighty personal matter.

Second, this weighty matter has always been conceptualized and practiced as an ordeal that learners need to accept and to overcome, as expressed in the following quote from Mencius that a Taiwanese father gave to his college-attending daughter who wanted to quit her physics assignment: "When heaven is about to confer a great responsibility on one, it will first test one's resolution, subject one's sinews and bones to hard work, expose one's body to hunger, put one to poverty, and place obstacles in the path of one's deeds, so as to stimulate one's mind [and heart], strengthen one's nature, and improve wherever one is incompetent."[57]

However, this ordeal is not conceived of as a punitive, draconian measure, intended to harm the learner; rather, it is to guide the learner along the learning path. The ordeal is not supposed to deter the learner; it serves instead to summon and to test the learner's resolve and commitment. In stepping up to the ordeal, the learner forges personal strength and displays it to his or her family and community. This commitment also helps the learner accomplish the actual learning. Thus, Chinese learners are expected to engage in learning with the anticipation and readiness to face challenges.

Challenges, or any situation that makes learning difficult for the learner, are expressed in Chinese as hardship, *ku* (苦). In Chinese, this term also means bitterness. But bitterness is not something that overwhelms the person. Learning is the ultimate good, but the process is wrought with endless challenges. To achieve the ultimate good, one must face and overcome any challenge and hardship. The virtue that enables one to face up to and combat each challenge is endurance of hardship (刻苦).

To gain a sense of the kind of challenges that loom in the learning process, we can look at the expressions I collected from my lexicon study: *Endurance of hardship* (刻苦); *study assiduously* (刻苦學習); *study diligently and practice hard* (勤學苦練); *take great pains to study* (苦心攻讀); *endurance of hardship is the boat to the boundless sea of learning* (學海無涯苦作舟); *without bitter coldness, where does the fragrance of the plum flower come from?* (沒有一番寒刺骨，哪得今日梅花香); *the sharp blade of the sword comes from constant grinding, the fragrance of plum flowers comes only after bitter coldness* (寶劍鋒從磨礪出，梅花香自苦寒來); *there is no difficulty on earth for the resolute person (*世上無難事，只怕有心人*); a ten-year cold window (study ten years)* (十年寒窗); *Sun Jing hung his hair on a beam overhead (to keep himself awake for studying)* (頭懸梁); *Su Qin stung his thighs with an awl (to keep himself awake for studying)* (錐刺股); *digging a hole in the wall to borrow the neighbor's light (for studying at night)* (鑿壁偷光); *Zhu Maicheng studied while wood-cutting* (負薪讀書).

This set of expressions contains two kinds of hardship. The first one is the learning process itself, which has two further challenges. One is the length of time required to learn anything well, as discussed under diligence/self-exertion. The other challenge is the routine encounter of inherently difficult knowledge or skill that is worth pursuing. For example, learners face not understanding a concept or a math proof, or not being able to perform a specific artistic act. Giving up is not an option, so the learner has to accept the challenge. The learner will either spend more time on a given concept or performance, or seek out the teacher and peers for help. If such extra work will not lead to desired results, the learner will endure more hardship to double his or her effort. Each time the learner has to do that, he or she will take time away from leisure and likely experience such learning as overcoming hardship. In my interviews with Chinese college students about what hardship meant to them in school, one student shared a common scenario of enduring hardship: "I often force myself not to go to dance parties at weekends. I, too, want to have fun, and I, too, want to join my friends. But I have to let that go because I have to study. It's really hard, really hard (真的是苦, 真苦)!" But when asked if he felt regret, he said no; he was instead proud of himself for being able to endure such hardship.

As noted earlier, Chinese learners exert themselves also to offset their perceived lack of natural ability/talent. In my ideal learner study, lack of inherent ability was identified as another common form of hardship that such learners must face and overcome. Instead of giving up, one will spend more time, seek more help, and practice more. They believe that for a given amount of knowledge or skill, one can learn quickly if one is lucky enough to be born with superior intelligence/talent. But if one is not so bright, there is no use to lament over what one was not given at birth. Instead, one simply needs to increase the length of time and the amount of effort. One should focus on what one *can* do and try to use that to offset one's weaknesses. With this kind of attitude and work, it is believed, everyone will reach the finish line eventually.[58]

However, the most prevalent hardship throughout Chinese history has been poverty that learners had to endure. The subjects of the last four expressions on the aforementioned list are the cultural exemplars who endured great hardship of poverty in order to learn. There were numerous others who also belong to the pantheon of such cultural icons. But these four are representative. Every schoolchild knows these stories because they are in the textbook and are very popular readings for beginner schoolchildren, such as the household book *Three Character Classic* (三字经). These stories are taught to children in order for them to emulate the very virtue of endurance of hardship as embodied by these exemplars. The four expressions that entered my lexicon list are the condensed forms of the specific ways of overcoming poverty that these people showed. Here are the highly abridged four stories found in *Three Character Classic*[59]:

> *Su Qin Stung His Thighs with an Awl* (錐刺股). Some time during the period of the Warring States (475–221 BCE), Su Qin was a very poor child. He had to do a lot of harsh labor to make a living. He had time only at night. But he kept falling asleep. One night he fell asleep at his desk but woke up by being accidentally scratched by an awl. He got this idea and used it to sting his leg whenever he got groggy. His family begged him not to do that again. But he said that he had to study. He studied all the books on politics and warfare and developed a new strategy for defense.

He persuaded six smaller states to form a union to defend themselves against the most powerful state, Qin. Su Qin became prime minister of the union that successfully deterred Qin's invasion for many years.

Digging a Hole in the Wall to Borrow the Neighbor's Light (鑿壁偷光). Kuang Heng, who lived some time during the Han Dynasty (202 BCE–220 CE), was also very poor. He learned the Chinese characters from a relative. During the day, he had to work in the fields. He studied only at night, but his family could not afford light. However, he was determined to learn. So he chiseled a hole in the wall to borrow the neighbor's light. This was how he read. Later he became prime minister of the Han Dynasty.

Zhu Maicheng Studied While Wood-Cutting (負薪讀書). Some time during the Han Dynasty, the poor boy Zhu Maicheng had to cut wood for a living, but never gave up on learning. Whenever he had a few minutes, he would study under a tree. Later, he put the book on top of his wood carrier and read while carrying wood on his shoulders. This was how he got his knowledge. Later he became a high official at the court.

Sun Jing Hung His Hair on a Beam Overhead (頭懸梁). Sun Jing, who lived some time during the Jin Dynasty (266–420 CE), had no time to study during the day because he had to do hard labor for a living. He could study only at night. He was afraid of falling asleep while reading his books at night, so he tied his hair with a string to the beam. When he dozed away, the string would pull his hair to awake him. This was how he gained his knowledge. Later Sun Jing became a famous Confucian scholar.

These learners were ingenious in inventing creative ways to study despite poverty, with some even using means not most conducive to their own well-being. Interestingly, historical records show that many family members could not bear this level of endurance of hardship and would rather that they not use such means. But these learners were determined and persisted. All of them had noted achievements in history. The message is clear that although poverty is very difficult to overcome, it is not a sufficient reason for giving up on learning. Additionally, endurance of hardship was indeed a strong theme in my ideal learner study. There was no reference to backing down because of poverty for any of the described ideal learners. The respondents showed much admiration and respect for these learners. This kind of admiration and respect is also extended to any such learners from any culture, for example, the young Abraham Lincoln reading by the candlelight and the courageous Frederick Douglass learning how to read as a slave boy.

Perseverance (恆心)

Perseverance and endurance of hardship are related virtues, but they have different emphases and therefore serve somewhat different psychological functions in the learning process, as noted in previous chapters. Endurance of hardship acknowledges the existence of hardship and the learner's need to overcome it. Perseverance emphasizes the completion of the whole learning process from the very beginning to the very end. This virtue is used to remove any challenge including, but not limited to, hardship, interruption (which may not be hardship, e.g., friends' invitation to have fun), disturbances, distractions, boredom, and setbacks. Perseverance is following through one's learning no matter what happens.

The expressions I collected for my lexicon study contain these items: *Perseverance (持之以恆); learning is like sailing against the current; it moves backward if not forward (學如逆水行舟，不進則退); utmost sincerity can break stone (精誠所致，金石為開); indefatigable (in learning) (孜孜不倦); study all night by the candlelight (秉燭天明); unremitting effort (不懈努力); and keep on carving unflaggingly, and metal and stone will be engraved (Xun Zi) (鍥而不舍，金石可鏤).* These expressions point out that there are obstacles in learning, as captured in a metaphor, sailing against the current. One will not achieve much if small obstacles can deter one from continuing. No matter what happens, the learner shall keep on learning. With such persistence, one will achieve big things in the end, hence breaking stone. If, however, one aborts one's effort halfway, one will accomplish little in learning.

Long-term practice that was discussed in the practice section goes hand in hand with perseverance. Each time the learner is frustrated about not making sufficient progress, the voices from the self, teachers, parents, and peers remind one to keep learning and practicing. This process is identical to that of a marathon runner who may want to quit at moments but mentally pushes oneself to continue. The cheering crowds also give the runner encouragement and strength to continue. As a result, the runner finishes. Here is an example of such learning displayed by a Taiwanese student studying at one American university:

> Shirley was doing her physics homework one night. After having worked hard for several hours, she could not continue any longer. Her father called from Taiwan asking how she was doing. "Dad," said Shirley, "I have been working hard on my physics homework, but I can't figure out this problem, and I am tired. I am done for the day." Upon hearing that his daughter was going to quit, the father recited Mencius' above-quoted passage on how a person shall face challenges and persist if this

> person is to assume great responsibilities in the future. Then the father asked Shirley to persist a bit longer. She did, until she solved her physics problem.

When discussing with me her family's support for her learning, Shirley expressed gratitude for her father's unflagging teaching of perseverance. Apparently, her father gave this kind of encouragement to her throughout the four years of Shirley's undergraduate education. Undoubtedly, Shirley understands what perseverance means.

One might ask what happens to children who quit. To be sure, learners quit all the time for all kinds of reasons. However, in school and at home, whenever a child shows signs of giving up, teachers, peers, and parents as well as the extended kin are mobilized to do what Shirley's father did: to reason with the child and to help the child improve. If the child does not have illness or another good reason, teachers and parents will show the child exemplary role models who persisted, overcame their inclinations to quit, and achieved their goals. Some of these models are historical figures, but others may be just peers in the same class. Chinese schools frequently hold a special meeting called "Exchange Study Methods" (交流學習方法), where peers share how they study and exercise various learning virtues. Children share obstacles they have encountered, particularly their own frustrations and difficulties, and specific strategies they used to deal with such problems. Hearing that one's peers also have these difficulties confirms that it is not unusual for one to experience similar problems. Children can learn from each other about how to improve one's perseverance (other virtues). When teachers see a given child make progress, there is often public praise for such improvement of virtue. Not only do such learning opportunities echo the importance of virtues, but they also provide specific ways to acquire and to exercise them in learning. These approaches in school help perpetuate learning virtues from generation to generation.

Concentration (專心)

The last, but of course not least, learning virtue is concentration. As noted in other chapters, concentration is particularly important in the actual learning process. If earnestness/sincerity is the overall virtue, diligence/self-exertion the temporal virtue, endurance of hardship is the hardship virtue, and perseverance the follow-through virtue, concentration is the virtue of each moment for each learning activity.

My lexicon study obtained these expressions: *Concentrate on learning (專心學習); devote oneself to studying (潛心學習); put one's heart into one's study (用心讀書); bury oneself in books (埋頭書齋); make great effort to* *study and research (努力鑽研); is engrossed in reading sacred books and oblivious to what happening in the world (兩耳不聞窗外事，一心只讀聖賢書); (so absorbed as to) forget food and sleep (廢寢忘食); and study with rapt attention (全神貫注地學習).* These expressions reveal a very high degree of concentration, attention, and wholehearted dedication to learning.

Chinese/East Asian cultures strongly emphasize contemplative learning – that is, the fact that the learner needs quiet time and space to study and to ponder. In classroom settings, children's undivided attention is demanded by the common learning topic at hand. Side talking with each other is not encouraged. Noisy environments with unrelated talking where the teacher is teaching and students studying are regarded as detrimental to learning. Talking aloud to oneself while studying is also believed not to be conducive to one's learning. In fact, the Neo-Confucians in the Song-Ming times borrowed from the Buddhist meditative practice to clear one's mind and promoted quiet sitting as a key method for concentration. Recent research by Kim[60] provides support that quiet learning is valued by Asian learners, and speaking while learning interferes with Asian learners' information processing (but it does not seem to interfere with that of European-American learners).

However, concentration is not an easy virtue to acquire. Chinese learners are keenly aware that much of anyone's struggle in learning is lack of concentration. This problem applies even to people who do not lack any motivation and conviction about learning. Today, neuroscience tells us that attention/concentration/focus is a manifestation of our executive functioning in the prefrontal lobe, which does not fully mature until late adolescence and early adulthood. Concentration is a clearly observable problem among young, normally developing children. And it is a tremendous challenge for children with the attention deficit and hyperactivity disorder (ADHD), a common neurological disorder affecting many children.

In order to help children, Chinese parents and teachers train children for concentration with a number of strategies. One of them is to create quiet space for children to sit down and to study without any disturbance. For example, teachers may ask a class of children to sit down and to count how long they can sit without fidgeting or moving away.[61] This training makes children sensitive not only to their own need for concentration, but also to how specifically they can monitor themselves. Another effective strategy is to have children practice skills that require slowing down and exhibiting fine motor control such as imitating Chinese calligraphy with exactitude (see Figure 4.3 for how a grandfather demonstrates rapt attention to his grandson in writing Chinese calligraphy), training for holding chopsticks,

FIGURE 4.3. A Chinese grandfather teaching his five-year-old grandson how to concentrate on writing Chinese characters with a brush. Courtesy of Heidi Fung.

and producing disciplined movements of martial arts, dance steps, and other athletic forms. A commonly observed classroom ritual is for the class monitor, a peer, to call the class to stand up as the teacher steps into the classroom. The class then greets the teacher in chorus: "Good day, Teacher!" The teacher returns the greeting "Good day, Students! Please sit down!" and "Let's begin class!" This ritual takes less than thirty seconds. Although short, it marks the end of the leisure time and the beginning of the serious and sacred class time to which all students' attention is directed and demanded.[62]

Parents also use various strategies to help improve their children's concentration. It is a common scenario in living compounds to hear parents instruct their children with the term concentration *zhuanxi* (專心) as "You need to be more *zhuanxi* in your studies" and "Are you not *zhuanxi* in school?" Our recent study with recorded mother–child conversations about learning in Taiwan indicates a much higher rate of exchanges about concentration than their European American mother–child pairs.[63] In spite of their very small living space, parents often clear off the dinner table after dinner and turn off the TV to ensure a quiet homework space and time for their children. When children are actually working, parents may stay nearby instead of leaving them to their own devices, engaging in a style of need-based intervention. When siblings distract each other, parents step in to remind them that they need to concentrate. This way, daily homework and review are done as expected because the time and space as well as the concentration are all ensured.[64] Some parents will even go a step further to help their children acquire and exercise the virtue of concentration. For example, when I was a child, our town had a semi-barter market system where farmers brought their produce and poultry to the market on a scheduled weekday to sell their goods and then to buy the goods they needed. When the day came, the whole town turned into a noisy, bustling market. My father used to put a stool on the street outside our courtyard and asked me to read a book without being disturbed by the market noise. I sat there and read my books until I was no longer distracted by the noise.

Those who demonstrate good concentration are not believed to have an inherent superior temperament or personality trait for concentration. Chinese learners believe strongly that anyone can acquire the virtue of concentration and everyone should. Students with good concentration are asked to share their learning experience and are admired. Other children are encouraged to emulate these models as well as finding ways to improve their own concentration.

The prominence of the Chinese virtue-oriented learning process received strong support from research conducted by others as well as myself. In my ideal learner study, for example, as many as 86 percent of the respondents mentioned one or more of these virtues. In comparison, only 38 percent of the European-American respondents wrote about virtue-oriented learning processes (but 96 percent mentioned mind-oriented learning processes). Another study on learning agency among current Chinese students from late elementary to high school also found that ordinary Chinese students in China endorse predominantly these learning virtues as underlying their daily learning processes.[65] Moreover, Chinese children as young as four start to internalize these learning virtues. The older children showed increasingly more such orientation.[66] Finally, the admiration that such displayed learning virtues engender in their teachers and peers is no less than the admiration expressed toward the exploratory, inquiring, critically thinking, and expressive spirit of Western learners.

NOTES

1. See DeCasper, A. J., & Spence, M. J. (1986). Prenatal maternal speech influences newborn's perceptions of speech sounds. *Infant Behavior and Development, 3*, 133–150 for a study on how fetal learning takes place. See also Brown, R. (1973). *A first language: The early stages*. Cambridge, MA: Harvard University Press for the documentation of lack of adult instructions for child language acquisition. See Rakoczy, H., Warneken, F., & Tomasello, M. (2008). The sources of normativity: Young children's awareness of the normative structure of games. *Developmental Psychology, 44*(3), 875–881 for research on young children's sense of social norms.

2. See Gardner, H. (1991). *The unschooled mind: How children think and how schools should teach.* New York: Basic Books for a review of research on how children learn before formal schooling. Human capacity to learn is most evident at birth. The fast mental growth in early infancy cannot be attributed to cultural influence because infants do not have enough time to acquire cultural knowledge. However, from the perspective of human evolutionary history, culture may have played an important role in its interaction with biology and thus may have jointly produced human learning capacity. See note 7 to Chapter 1 for more discussion on this process.
3. Smits, W., & Stromback, T. (2001). *The economics of the apprenticeship system.* Cheltenham: Elgar.
4. See Olson, D. R. (1994). *The world on paper: The conceptual and cognitive implications of writing and reading.* New York: Cambridge University Press for human literacy development. See Damerow, P. (1998). Prehistory and cognitive development. In J. Langer & M. Killen (Eds.), *Piaget, evolution, and development* (pp. 227–270). Mahwah, NJ: Erlbaum for earliest formal schooling in the Middle East; and Siu, M. K. (2004). Official curriculum in mathematics in ancient China: How did Candidates study for the examination? In L.-H. Fan, N.-Y. Wong, J.-F Cai, S.-Q. Li, & T.-Y. Tso (Eds.), *How Chinese learn mathematics: Perspectives from insiders* (pp. 157–185). Singapore: World Scientific for an account of early Chinese formal schooling. In the Chinese system, from the seventh century to 1905, all people in principle could participate in the Civil Service Examination. It is important to point out that only those who could afford education benefited from this system.
5. To be sure, ample research shows that just because compulsory education exists in the world does not mean that all children receive equal education. Large inequity exists between middle-class and low-income populations across the globe. The United States is a good example. Still, compulsory education is an important achievement in human history. See the first reference of note 1 to Chapter 3 for more discussions.
6. All of the terms/expressions in English and Chinese that I present in this chapter came from the study that is referenced in note 54 to Chapter 3.
7. The statistical analysis of the sorted lexicon placed the most frequently grouped items to the far left, the second most frequently grouped items next, and so forth. Thus, the order in which categories emerged indicates relative magnitude of group consensus of the category. The further left the cluster, the heavier the magnitude. See Figure 3.1 in Chapter 3.
8. Edwards, C., Gandini, L., & Forman, G. (Eds.). (1998). *The hundred languages of children: The Reggio Emilia approach – advanced reflections.* (2nd ed.). Westport, CT: Ablex.
9. Pintrich, P. R., Smith, D. A. F., Garcia, T., & McKeachie, W. J. (1993). Reliability and predictive validity of the motivated strategies for learning questionnaire (MSLQ). *Educational and Psychological Measurement, 53*, 801–813; and Purdie, N., & Hattie, J. (1996). Cultural differences in the use of strategies for self-regulated learning. *American Educational Research Journal, 33*, 845–871.
10. See Hidi, S., & Renninger, K. A. (2006). The four-phase model of interest development. *Educational Psychologist, 41*(2), 111–127 for research on interest and its development in learning.

11. See Cole, M., Cole, S. R., & Lightfoot, C. (2005). *The development of children.* (5th ed.). New York: Worth Publishers for basic knowledge on child development.
12. See the first reference in note 33 to Chapter 3. Gardner and Winner did not appreciate Chinese adults' interference with their son's exploration and concluded quite observantly that this Chinese adult behavior may reveal some fundamentally different beliefs about child development and learning.
13. See note 8 to this chapter.
14. Kuhn, D. (1991). *Education for thinking*. Cambridge, MA: Harvard University Press.
15. Nobel Museum (n.d.). Retrieved October 5, 2010 from http://nobelprize.org/nobel_prizes/physics/laureates/
16. See note 15 to this chapter. Also see Gardner, H., Csikszentmihalyi, M., & Damon, W. (2001). *Good work: When excellence and ethics meet*. New York: Basic Books for the testimonies of current scientists in the biomedical field and genetics on their "thrill of scientific inquiry" (p. 73).
17. Perkins, D. N. (1995). *Smart schools.* New York: Free Press; also see the second reference in note 16 to this chapter for the essential role of "quality of thinking" (p. 74) and "rational thinking" (p. 75) in scientific research.
18. See McBride, R. E., Xiang, P., Wittenburg, D., & Shen, J.-H. (2002). An analysis of preservice teachers' dispositions toward critical thinking: A cross-cultural perspective. *Asian-Pacific Journal of Teacher Education, 30*(2), 131–140 for a review of the importance of critical thinking in the West.
19. Ennis, R. (1987). A taxonomy of critical thinking dispositions and abilities. In J. Baron & R. Sternberg (Eds.), *Teaching thinking skills: Theory and practice* (pp. 9–26). New York: Freeman, p. 10.
20. Facione, P. (1990). *Critical thinking: A statement of expert consensus for purposes of educational assessment and instruction. Executive summary of "The Delphi Report."* Milbrae, CA: The California Academic Press.
21. Nobel Museum (n.d.). Retrieved October 5, 2010 from http://nobelprize.org/nobel_prizes/physics/laureates/
22. See note 21 to this chapter.
23. See note 18 to Chapter 2, p. 11.
24. A fuller account of this tradition will be presented in Chapter 8 in contrast to the reluctance to speak in East Asia. See Sternberg, R. J. (1985). Implicit theories of intelligence, creativity, and wisdom. *Journal of Personality and Social Psychology, 49*, 607–627 for his research on verbal ability as an important component of the Western conception of intelligence.
25. See Wang, Q. (2004). The emergence of cultural self-constructs: Autobiographical memory and self-description in European American and Chinese children. *Developmental Psychology, 40*(1), 3–15; and Wang, Q., & Leichtman, M. D. (2000). Same beginnings, different stories: A comparison of American and Chinese children's narratives. *Child Development, 71*, 1329–1346; and also the first reference in note 6 to Chapter 3 for differences in children's narratives.
26. See note 8 to this chapter.
27. Goleman, D. (1995). *Emotional intelligence: Why it can matter more than IQ*. New York: Bantam.
28. See note 54 to Chapter 3.

29. See notes 26 and 27 to Chapter 3 for detailed descriptions of Japanese preschool and elementary school classrooms. See Hsueh, Y. (1999). *A day at Little Stars Preschool, Beijing, China*. Unpublished video footage; Wang, T., & Murphy, J. (2004). An examination of coherence in a Chinese mathematics classroom. In L.-H. Fan, N.-Y. Wong, J.-F Cai, S.-Q. Li, & T.-Y. Tso (Eds.), *How Chinese learn mathematics: Perspectives from insiders* (pp. 107–123). Singapore: World Scientific; and Cortazzi, M., & Jin, L.-X. (2001). Large classes in China: "Good" teachers and interaction. In D. A. Watkins & J. B. Biggs (Eds.), *Teaching the Chinese learner: Psychological and pedagogical perspectives* (pp. 115–134). Hong Kong: Comparative Education Research Centre for documentation of how children show earnestness/sincerity in classroom learning.
30. Paine, L. W. (1990). The teacher as virtuoso: A Chinese model for teaching. *The Teachers College Record, 92*(1), 49–81.
31. See notes 21, 26, 27, and 50 to Chapter 3.
32. Levy, R. I. (1973). *Tahitians*. Chicago: University of Chicago Press.
33. See notes 43 and 44 to Chapter 3. Also see Marton, F., Dall'Alba, G., & Beaty, E. (1993). Conceptions of learning. *International Journal of Educational Research, 19*, 277–300 for their research on the role of memorization and understanding among Chinese learners.
34. Zhang, D.-Z., Li, S.-Q., & Tang, R.-F. (2004). The "two basics": Mathematics teaching and learning in Mainland China. In L.-H. Fan, N.-Y. Wong, J.-F Cai, S.-Q. Li, & T.-Y. Tso (Eds.), *How Chinese learn mathematics: Perspectives from insiders* (pp. 189–207). Singapore: World Scientific.
35. See the second reference in note 8 to Chapter 3. Also see two new books published recently: Colvin, G. (2008). *Talent is overrated: What really separates world-class performers from everybody else*. New York: Portfolio; and Coyle, D. (2009). *The talent code: Greatness isn't born. It's grown. Here's how*. New York: Bantam Dell for how great performances from the West also depend on this type of "deliberate" or "deep" practice.
36. Chen, C., & Stevenson, H. W. (1989). Homework: A cross-cultural examination. *Child Development, 60*, 551–561; Helmke, A., & Tuyet, V. T. A. (1999). Do Asian and Western students learn in a different way? An empirical study on motivation, study time, and learning strategies of German and Vietnamese university students. *Asian Pacific Journal of Education, 19*(2), 30–44; Peng, S. S., & Wright, D. (1994). Explanation of academic achievement of Asian American students. *Journal of Educational Research, 87*(6), 346–352; third reference in note 31 to this chapter; and Purdie, N., Hattie, J., & Douglas, G. (1996). Student conceptions of learning and their use of self-regulated learning strategies: A cross-cultural comparison. *Journal of Educational Psychology, 88*, 87–100.
37. See note 44 to Chapter 3.
38. Li, J. (1998). The power of embedding learning beliefs in everyday learning: Language arts texts as a vehicle for enculturation. Unpublished manuscript. I tallied all content in China's elementary school's language arts curriculum published in 1997. I found that 36% of all genres (e.g., stories, excerpts of novels, essays, songs, poems, letters, etc.) contained learning beliefs (e.g., moral self-perfection and contribution to society) and learning virtues (e.g., Madam Currie worked long hours in the lab).

39. See the first reference in note 61 to Chapter 3.
40. See second reference in note 8 to Chapter 3.
41. See in note 5 to Chapter 2, 2.11, p. 78.
42. See the second reference in note 36 to this chapter.
43. See note 21 to Chapter 3.
44. See notes 26 and 28 to Chapter 3.
45. See the first reference in note 61 to Chapter 3.
46. See the first reference in note 61 to Chapter 3. See Huang, H.-M. (2004). Effects of cram schools on children's mathematics learning. In L.-H. Fan, N.-Y. Wong, J.-F Cai, S.-Q. Li, & T.-Y. Tso (Eds.), *How Chinese learn mathematics: Perspectives from insiders* (pp. 282–306). Singapore: World Scientific for how, among other reasons, putting in more time and effort by attending the so-called cram schools in Taiwan is believed to compensate for children's lack of ability.
47. See the first reference in note 61 to Chapter 3.
48. See the third reference in note 4 to this chapter, and Li, J.-H. (2004). Thorough understanding of the textbook: A significant feature of [the] Chinese teacher manual. In L.-H. Fan, N.-Y. Wong, J.-F Cai, S.-Q. Li, & T.-Y. Tso (Eds.), *How Chinese learn mathematics: Perspectives from insiders* (pp. 262–281). Singapore: World Scientific.
49. Anderson, G. I. A. (Ed.). (1977). *Masterpieces of the Orient* (6th ed.). New York: Norton, pp. 421–422.
50. See references in note 75 to Chapter 2 and Kember, D. (2000). Misconceptions about the learning approaches, motivation, and study approaches of Asian students. *Higher Education, 40*, 99–121 for criticism of Chinese rote learning. Many Chinese educators themselves also join in this condemnation.
51. Geertz, C. (1973). *The interpretation of culture*. New York: Basic Books.
52. Bertensson, S., Leyda, J., & Satina, S. (2002). *Sergei Rachmaninoff: A lifetime in music*. Bloomington: Indiana University Press, pp. 10–11.
53. Marcovitch, S., & Zelazo, P. D. (2009). A hierarchical competing systems model of the emergence and early development of executive function. *Developmental Science, 12*(1), 1–18.
54. See note 44 to Chapter 3 and the second reference in note 33 to this chapter, p. 81.
55. See the second reference in note 36 to this chapter.
56. Chao, R. K. (1996). Chinese and European American mothers' views about the role of parenting in children's school success. *Journal of Cross-Cultural Psychology, 27*, 403–423.
57. See the second reference in note 71 to Chapter 3, p. 330. The original translation by D. C. Lau (note 50 to Chapter 2) was done with the English masculine pronouns *he*, *his*, and *him* for the Chinese term 人, which is not gendered. Although before the twentieth century, few women were ever educated and involved in political life, I maintain that the pronouns for ancient texts should be translated with the gender-neutral pronouns *one* and *one's*. This insistence is especially relevant given that the father quoted Mencius' passage to encourage his daughter.
58. This is probably why the Aesop's fable *Hare and Tortoise* (from its hundreds of fables) enjoys eternal popularity in Chinese/Asian cultures. The opportunity

to learn a skill over a very long period of time where these Chinese learning virtues are learned, applied, and preserved is likely to be a truer reason why so many Chinese/Asians parents enroll their children in learning how to play Western classical music instruments. This penchant of Asian parents stands in sharp contrast to Western classical music being viewed as "museum music" by pop musicians in the postmodern Era.

59. See note 77 to Chapter 2.
60. Kim, H. S. (2002). We talk, therefore we think? A Cultural analysis of the effect of talking on thinking. *Journal of Personality and Social Psychology, 83*, 828–842.
61. See note 26 to Chapter 3.
62. See the fifth reference in note 29 to this chapter and note 30 to this chapter.
63. Li, J., Fung, H., Liang, C.-H., Resch, J., Luo, L., & Lou, L. (2008, July). When my child doesn't learn well: European American and Taiwanese mothers talking to their children about their children's learning weaknesses. In J. Li & H. Fung (Chairs), *Diverse paths and forms of family socialization: Cultural and ethnic influences*. Symposium paper presented at a biannual conference of the International Society for the Study of Behavioral Development, Würzburg, Germany.
64. See the second reference in note 4 to Chapter 3.
65. Li, J. (2006). Self in learning: Chinese adolescents' goals and sense of agency. *Child Development, 77*(2), 482–501.
66. See the first reference in note 6 to Chapter 3.